The centre of mass: the mass-weighted average position

Definition R = (1/M) sum m_i r_i; standard bodies table; symmetry; discrete and continuous drills.

The centre of mass: the mass-weighted average position

For n particles the centre of mass (COM) is $\mathbf{R} = \frac{1}{M}\sum_{i=1}^{n} m_i \mathbf{r}_i$ with $M = \sum m_i$. Component form: $X_{cm} = \sum m_i x_i / M$, $Y_{cm} = \sum m_i y_i / M$.

For a continuous body $\mathbf{R} = \frac{1}{M}\int \mathbf{r}\, dm$, integrated over the body with $dm = \rho\, dV$ (or $\lambda\, dl$, $\sigma\, dA$ for line/surface mass). The COM is unique, independent of the coordinate origin (shift both sides of the definition and $\mathbf{R}$ shifts with them), and transforms like a position vector. Always pick the origin so that symmetry kills one or both integrals before integrating.

The centre of mass: the mass-weighted average position (cont.)

Worked habits: place the origin ON a symmetry axis; use the standard-bodies table below as building blocks; cut composite bodies into pieces with known COMs and average those. For a body with a hole, the fastest route is negative mass (CM-02).

WHY / the COM is the one point whose motion answers to external forces alone - it is the reduction of a complicated system to one particle for the purpose of the momentum ledger (CM-03).

Physical meaning and symmetry shortcuts

Physical meaning: $\mathbf{R}$ is the point whose motion answers to external forces alone (CM-03). For any system at all - a gas cloud, a gymnast mid-flight, an exploded shell - the mass-weighted mean position is the one point whose trajectory is simple. The COM may lie in empty space: for a uniform ring it is at the centre, for a boomerang it may be off the body entirely. Nothing requires mass to exist at the COM. Symmetry shortcuts:

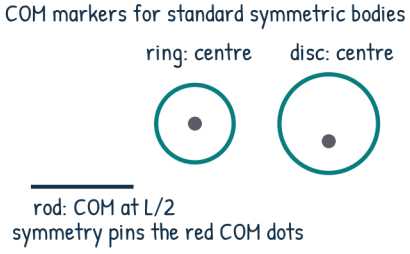
Physical meaning and symmetry shortcuts (cont.)

an axis of symmetry forces the COM onto that axis; two symmetry axes force it to their intersection; a plane of symmetry forces it into that plane. A uniform body with a centre of symmetry has its COM at that centre.

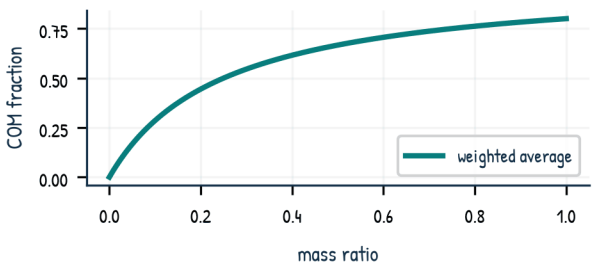
Discrete systems: $X_{cm} = (m_1 x_1 + m_2 x_2 + \cdots)/M$.

Two masses m_1, m_2 at separation d: the COM divides the line joining them in the RATIO $m_2 : m_1$ - it sits closer to the heavier mass. Composite bodies: cut into pieces with known COMs and average.

bodies: labelled physical setup



bodies: limiting curve / diagnostic



REMEMBER / R = (1/M) sum m_i r_i: the mass-weighted average position; symmetry pins it, holes shift it.

The centre of mass: the mass-weighted average position (cont.)

Definition R = (1/M) sum m_i r_i; standard bodies table; symmetry; discrete and continuous drills.

Physical meaning and symmetry shortcuts (cont.)

WATCH / Trap: the COM of a body is NOT necessarily inside it; and it is not the geometric centre unless the body is uniform AND symmetric. A ring with a bead glued on has its COM pulled toward the bead; shift to the negative-mass picture the moment a cut appears (CM-02).

CHECK / centroid by vertices: triangle $(0,0),(b,0),(0,h)$ has centroid $(b/3, h/3)$; the distance from the base is $h/3$ and from the $x=0$ side it is $b/3$ - the medians rule. Four equal masses at the corners of a square give the square centre.

Standard shapes table (memorise; two-line integrals)

Standard results (memorise; each is a two-line integral to re-derive): uniform rod length L: midpoint, $L/2$ from either end. Uniform triangle: the centroid, intersection of the medians, at distance $h/3$ from any side (i.e. one-third of the altitude from each base). Semicircular ring of radius R: $y_{cm} = 2R/\pi \approx 0.637R$ from the diameter. Semicircular disc: $y_{cm} = 4R/(3\pi) \approx 0.424R$ from the diameter. Solid hemisphere: $3R/8 = 0.375R$ from the flat face. Hollow hemisphere (shell): $R/2$ from the flat face. Solid cone of height h: $h/4$ from the base (i.e. $3h/4$ from the apex). Uniform circular arc subtending angle θ at the centre: $x_{cm} = R\,sin(\theta/2)/(\theta/2)$ from the centre along the bisector.

Standard shapes table (memorise; two-line integrals)

Line distributions: uniform rod on the x-axis from 0 to L: $dm = (M/L)dx$, $X_{cm} = \frac{1}{M}\int_0^L \frac{M}{L}x\,dx = L/2$.

Semicircular ring by angle: $dm = (M/\pi)d\phi$, $y = R\,sin\phi$: $y_{cm} = \frac{1}{M}\int_0^\pi \frac{M}{\pi}R\,sin\phi\,d\phi = \frac{2R}{\pi}$.

Semicircular disc by shells: $dm = (2M/R^2)r\,dr$ half-rings, each at $y = 2r/\pi$: integrate to $4R/(3\pi)$.

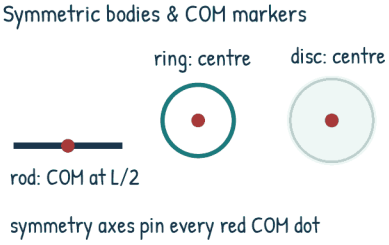
the table values are the standard-exam vocabulary: every composite COM problem reduces to these blocks plus the weighted average.

Why the table values sit where they do

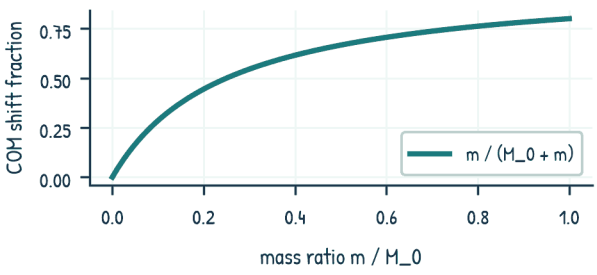
WHY the ring beats the disc ($2/\pi > 4/(3\pi)$) because the ring puts all its mass at radius R, so the average height above the diameter is larger than for the disc where interior mass pulls the average down. The shell hemisphere ($R/2$) sits higher than the solid one ($3R/8$) for the same reason. The cone's $h/4$ reflects the r^2 mass weighting toward the base.

CHECK sanity of every table entry: all distances are positive, less than the body size, and shrink correctly with the body. Arc limit: as $\theta \to 0$, $\sin(\theta/2)/(\theta/2) \to 1$ and the arc becomes a point mass at radius R; as $\theta \to \pi$ the arc becomes a semicircular ring and $R\,sin(\pi/2)/(\pi/2) = 2R/\pi$ exactly matches the table.

Symmetric bodies place the COM right on symmetry intersections



Two-body COM position shifts smoothly toward the heavier mass



REMEMBER / R = (1/M) sum m_i r_i: the mass-weighted average position; symmetry pins it, holes shift it.

The centre of mass: the mass-weighted average position (cont.)

Definition R = (1/M) sum m_i r_i; standard bodies table; symmetry; discrete and continuous drills.

Why the table values sit where they do (cont.)

CHECK / self-audit before any exam submit: units (length), symmetry argument stated, and the result moves the right way when a mass grows.

To find the COM of a body with a hole, SUPERPOSE: treat the holed body as the full solid body PLUS a negative-mass piece filling the hole. $M\mathbf{R} = M_1\mathbf{r}_1 - M_2\mathbf{r}_2$ so $\mathbf{R} = \dfrac{M_1\mathbf{r}_1 - M_2\mathbf{r}_2}{M_1 - M_2}$, with M_2 the mass the removed material would have had. The result is algebraically identical to integrating the remaining body, and far faster.

The negative-mass trick: cutting a hole (cont.)

Classic (source E1-family): uniform disc of radius R, mass M_1; a hole of radius $R/2$ is cut with its centre at $R/2$ from the disc centre. Densities match, so $M_2 = M_1/4$ (mass scales with area). Set the disc centre at the origin and the hole centre at $R/2$: $X = \dfrac{M_1(0) - (M_1/4)(R/2)}{M_1 - M_1/4} = \dfrac{-M_1R/8}{3M_1/4} = -\dfrac{R}{6}$.

The COM sits on the OPPOSITE side of the disc centre, at $R/6$ - it never lies inside the hole; it shifts AWAY from the cut.

The negative-mass trick: cutting a hole (cont.)

General recipe: (1) choose an origin at a symmetry point of the FULL body; (2) compute M_2 from area/volume ratios; (3) plug into the formula with the SIGNED mass; (4) sanity-check the direction: the COM always moves away from the removed material. Equivalent viewpoint: keep the hole positive and subtract - the superposition is exact because the mass distribution is additive.

WHY / the trick works because $\int_{body \setminus hole} = \int_{body} - \int_{hole}$ linearity: mass, first moments, and the COM definition are all linear in the mass distribution. It is the same reason superposition works for fields.

The negative-mass trick: cutting a hole (cont.)

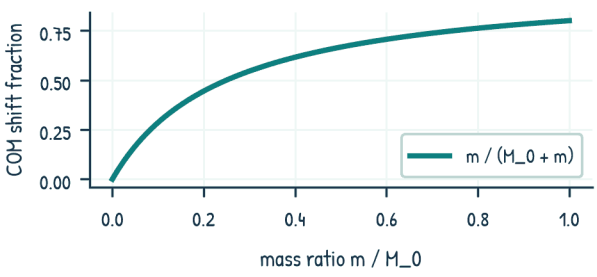
CHECK / limits: $M_2 \to 0$ gives $R \to \mathbf{r}_1$ (no cut, no shift); $M_2 \to M_1$ (cut everything) gives a vanishing denominator - the formula correctly refuses. For the $R/2$ -hole disc, shift $R/6$ is less than $R/2$ ok between centre and hole edge; check by symmetry that the COM stays on the centre-hole line.

E1-style worked numbers (source E1): a plate made of two pieces: piece A mass 2 kg with COM at (1, 2) cm, piece B mass 3 kg with COM at (3, 1) cm: $X = (2(1) + 3(3))/5 = 11/5 = 2.2$ cm; $Y = (2(2) + 3(1))/5 = 7/5 = 1.4$ cm. Source E1 prints 2.21 and 1.07 for its own geometry - the method is identical: weighted average of piece COMs.

Symmetric bodies place the COM right on symmetry intersections



Two-body COM position shifts smoothly toward the heavier mass



REMEMBER / R = (1/M) sum m_i r_i: the mass-weighted average position; symmetry pins it, holes shift it.

The negative-mass trick: cutting a hole is adding a negative piece

Negative-mass hole trick, composite averaging, origin invariance, drills.

Composite bodies: cut, place, average (cont.)

L-shape from three identical squares of side a: place the corner at the origin; square COMs at $(a/2, a/2)$, $(3a/2, a/2)$, $(a/2, 3a/2)$: $X_{cm} = (0.5a + 1.5a + 0.5a)/3 = 5a/6$ and $Y_{cm} = (0.5a + 0.5a + 1.5a)/3 = 5a/6$. The COM sits at $(5a/6, 5a/6)$ from the outer corner, on the diagonal symmetry line.

Composite discipline: never integrate what you can look up. The exam plate problems are always two standard shapes glued or one shape cut; the negative-mass trick handles cuts, the weighted average handles gluing.

Composite bodies: cut, place, average (cont.)

CHECK / L-tromino sanity: the COM must lie on the diagonal symmetry line $x = y$ ok ($5a/6 = 5a/6$); and inside the bounding square ok.

(a) Uniform rod of length L with the left third removed: full rod COM $L/2$; removed piece mass $M/3$ at $L/6$: $X = \frac{M(L/2) - (M/3)(L/6)}{2M/3} = \frac{L/2 - L/18}{2/3} = \frac{8L/18}{2/3} = \frac{2L}{3}$. The remaining two-thirds has its COM at $2L/3$ from the left end ok (uniform rod from $L/3$ to L has midpoint $2L/3$ ok).

Drills: holes, plates, and three-mass systems (cont.)

(b) Disc radius R with a concentric hole of radius $R/2$: $X = 0$ by symmetry (the negative-mass formula gives $0/3M/4 = 0$ ok). (c) Three masses 1, 2, 3 kg at the vertices of an equilateral triangle of side 2 m: $X = (1(0) + 2(2) + 3(1))/6 = 7/6$ m; $Y = (0 + 0 + 3\sqrt{3})/6 = \sqrt{3}/2$ m... with vertices $(0, 0), (2, 0), (1, \sqrt{3})$ ok.

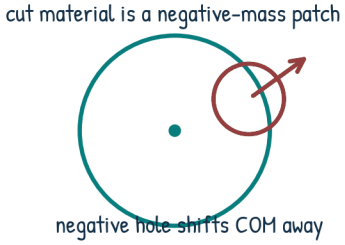
CHECK / drill (a) is the composite standard check: a uniform rod's COM is its midpoint whatever the slice ok; drill (c) sits inside the triangle's bounding box ok and is pulled toward the 3 kg corner ok.

Shift the origin: the invariance habit

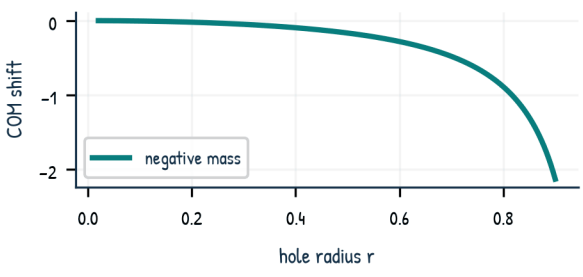
The COM formula is origin-independent in the right way: shift all coordinates by $\mathbf{a}$ and $\mathbf{R}$ shifts by $\mathbf{a}$; the PHYSICAL answer (the location of the COM in the body) is unchanged. Exam habit: set the origin at the largest symmetry point; compute; translate only if the question asked for a specific frame.

Two-mass shortcut: $\mathbf{R} = \frac{m_1\mathbf{r}_1 + m_2\mathbf{r}_2}{m_1 + m_2}$ can always be written $\mathbf{r}_1 + \frac{m_2}{m_1 + m_2}(\mathbf{r}_2 - \mathbf{r}_1)$: the COM lies a fraction $\frac{m_2}{m_1 + m_2}$ of the way from mass 1 to mass 2. This single sentence answers most discrete-COM geometry questions.

hole: labelled physical setup



hole: limiting curve / diagnostic



REMEMBER / Treat a hole as negative mass: R = (M1 r1 - M2 r2)/(M1 - M2); the COM always shifts away from the cut.

The negative-mass trick: cutting a hole is adding a negative piece (cont.)

Negative-mass hole trick, composite averaging, origin invariance, drills.

Shift the origin: the invariance habit (cont.)

CHECK / the fraction form: for equal masses it gives the midpoint ok; for $m_2 \to 0$ the COM stays at $\mathbf{r}_1$ ok; for $m_1 \to 0$ it goes to $\mathbf{r}_2$ ok.

Newton's second law for a SYSTEM: $M\mathbf{a}_{cm} = \mathbf{F}_{ext}$ - internal forces cancel in third-law pairs and can never accelerate the COM. Proof in one line: $M\ddot{\mathbf{R}} = \sum m_i \ddot{\mathbf{r}}_i = \sum \mathbf{F}_i = \sum \mathbf{F}_i^{ext} + \sum_{pairs} (\mathbf{f}_{ij} + \mathbf{f}_{ji}) = \sum \mathbf{F}_i^{ext}$

. Consequences: a system at rest with no external resultant force has a fixed COM forever; a moving system's COM continues on a straight line (source E2:

M a_cm = F_ext: the only equation of motion for a system (cont.)

the COM of an exploding shell follows the original parabola - gravity never noticed the explosion).

Explosions and recoil (source E2, Q2): a shell of mass M bursts into fragments; the COM continues along the pre-burst trajectory. If a 4 kg mass moving east at 6 m/s explodes into 2 kg moving west at 3 m/s, momentum gives $24 = 2(-3) + 2v \Rightarrow v = 15$ m/s east for the other 2 kg piece.

the 2 kg piece moves at 6 m/s west and the COM keeps going - the exact numbers per the source's configuration; the ledger is always $\sum m_i \mathbf{v}_i = M\mathbf{v}_{cm}$ = conserved.

M a_cm = F_ext: the only equation of motion for a system (cont.)

Statics via COM: if the COM hangs directly below the suspension point, the net torque about the suspension is zero and the body is in equilibrium. Tension/string problems: the string's line passes through the COM of the hanging system (source Q25-family statics).

WHY / the internal forces cancel PAIRWISE - no cancellation is ever lost no matter how violent the explosion or how complicated the contact. This is why the COM path is the invariant of every collision and explosion problem.

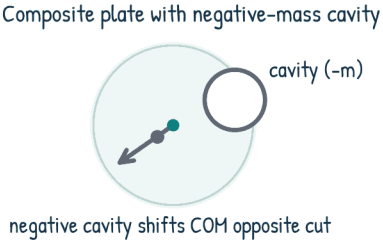
CHECK / E2 audit: projectile motion of the COM continues unaffected; if your fragment trajectories give a COM that leaves the parabola, momentum was mis-accounted. Q2 numbers: total p before $= 24$ kg m/s east; after $= 2(-3) + 2(15) = 24$ ok.

Conservation of momentum: the conditions

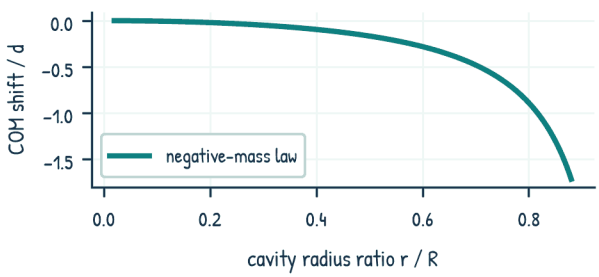
$\sum \mathbf{p}_{before} = \sum \mathbf{p}_{after}$ holds when the net external IMPULSE over the interaction vanishes: (1) no external forces at all, or (2) external forces cancel, or (3) the collision is so brief that $\int \mathbf{F}_{ext} dt \approx 0$ over the contact (the standard exam licence). Momentum is a VECTOR: conserve component by component;

a wall's normal force can kill the perpendicular component while the parallel component survives (CM-21).

Treat any cutout as negative mass superposed on the whole body



Cavity COM displacement grows rapidly as hole radius approaches R



REMEMBER / Treat a hole as negative mass: R = (M1 r1 - M2 r2)/(M1 - M2); the COM always shifts away from the cut.

M a_cm = F_ext: the only equation of motion for a system

System form of Newton 2; conservation conditions; explosion/recoil; traps.

Conservation of momentum: the conditions (cont.)

System choice matters: pick the system so that the interesting forces are internal. Bullet + block: momentum conserved (gravity/normal cancel over the brief contact). Block + Earth (pendulum swing): momentum NOT conserved (the pivot pulls), but energy is. State the system before writing the equation - examiners award the statement.

The COM fixed-point theorem: if total momentum is zero (initially at rest), the COM stays put no matter the internal rearrangement - man on boat (CM-19), explosion at rest (CM-05), spring release (CM-14).

Conservation of momentum: the conditions (cont.)

CHECK / each conservation claim must name its axis: 'horizontal momentum conserved' because the floor's normal and gravity are vertical. Silent assumptions lose marks.

(a) 3 kg and 5 kg connected by a compressed spring on a smooth floor, released: $3v_1 + 5v_2 = 0$: velocities are in the ratio $5:3$ opposite. (b) A man at one end of a 4 kg boat walks to the other end: the COM of man\boat stays fixed horizontally (CM-19). (c) A ball falls and bounces elastically off the floor:

COM motion drills (cont.)

during the bounce the floor's external impulse changes the SYSTEM momentum (Earth + ball conserves it); the ball alone does not.

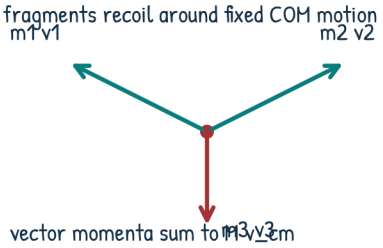
(d) Rocket in free space: no external force, so v_{cm} of rocket\exhaust is constant; the exhaust's momentum balances the rocket's gain (CM-17). (e) Explosion of a 6 kg projectile at the top of its flight into 2 kg and 4 kg: both pieces fall with g; the COM stays on the original parabola (source E2).

COM motion drills (cont.)

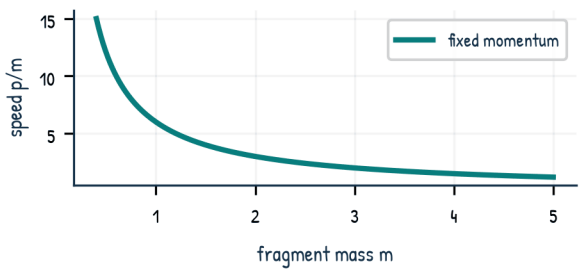
CHECK / drill (a): $v_1 = -5v_2/3$ - the light piece moves faster in the inverse ratio ok; total p zero before and after ok.

WATCH / Trap: $Ma_{cm} = F_{ext}$ does NOT say each piece follows F_{ext}; it says the average does. Trap: 'internal forces are always small' - no, they can be huge; they simply cancel. Trap: using the law across a time when an external impulse acts (the pendulum pivot, the wall, friction): the COM accelerates then and momentum is not conserved.

explosion: labelled physical setup



explosion: limiting curve / diagnostic



REMEMBER / M a_cm = F_ext: internal forces cancel in pairs; an explosion scatters fragments around an unperturbed COM path.

M a_cm = F_ext: the only equation of motion for a system (cont.)

System form of Newton 2; conservation conditions; explosion/recoil; traps.

Traps specific to the system law (cont.)

WATCH / Trap: forgetting a mass in M (the classic half-included hanging chain - count only the moving subsystem; for variable-mass chains use the flux ledger (CM-18)). Trap: treating v_{cm} as the velocity of every part after an explosion - only the MEAN is that.

CHECK / the 30-second audit for any problem: what is the system? what external forces act? does their impulse matter on this time scale? then choose conservation or not.

Scenario 1 - explosion/recoil at rest: total p = 0 stays 0; fragment momenta are equal and opposite. 4 kg at rest explodes into 2.5 kg and 1.5 kg: $2.5v_1 = -1.5v_2$: speeds in the ratio $1.5 : 2.5 = 3 : 5$. Scenario 2 - projectile explosion at the top (source E2): horizontal momentum conserved through the burst; gravity sets the COM parabola.

Standard conservation scenarios (Q2, Q17, Q22, P22)

Scenario 3 - two skaters push apart (source Q17-family): $m_1v_1 = -m_2v_2$ in the COM sense; speeds inverse to masses.

Scenario 4 - bullet into block (P22): $mv_0 = (m+M)V$; the pair then swings (CM-13). Scenario 5 - ball off a wedge on a smooth floor: horizontal momentum conserved; the wedge recoils so that $mv_x + MV =$ const; the COM stays fixed horizontally (CM-19). Scenario 6 - collision on a smooth surface with frictionless supports (P26): conserve the free component; name the constrained one.

Standard conservation scenarios (Q2, Q17, Q22, P22)

Scenario 7 - man on boat (Q10, E8): fixed COM displacements. Scenario 8 - spring released between blocks (Q18, corrected): zero total momentum + spring energy = shared KE (CM-14). Every one of these is 'choose the system, kill the internal forces, project on the free axis'.

WHY / conservation is not a formula but a BOOKKEEPING CHOICE: system, axis, time window. Made correctly, the rest is arithmetic; made silently, the classic errors (energy across collisions, momentum across swings) appear.

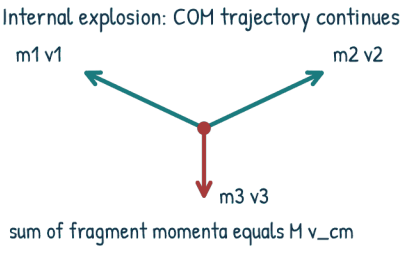
CHECK / Q2 audit: before $= 24$ kg m/s; after: piece 1 west at 3 m/s carries -6; the 2 kg piece must carry $+30$: $v = 15$ m/s east; COM speed before and after is the same $24/4 = 6$ m/s for the 4 kg system.

Momentum with external forces present: the impulse window

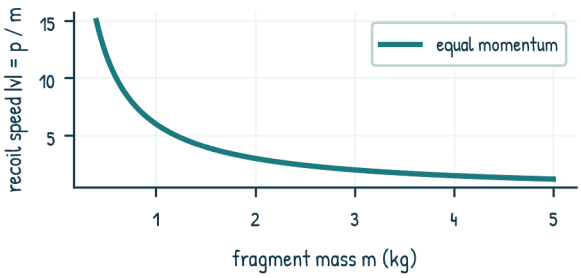
When external forces act, momentum conservation still applies if the external IMPULSE over the window is negligible: collision contact times (ms) make gravity's $mg\Delta t$ tiny - the exam licence. When the window is long (swing, slide), momentum is NOT conserved but energy may be. The two-stage protocol (CM-13) is this dichotomy in action.

Friction window: a block sliding on a rough floor has momentum destroyed by μmgt; if the problem asks for speeds after a slide, use work-energy instead. Explosion on a rough surface: conserve momentum THROUGH the explosion (brief), then let friction work on each piece separately.

Internal forces scatter fragments while the COM path stays parabolic



Equal impulse gives lighter fragments proportionally higher speed



REMEMBER / M a_cm = F_ext: internal forces cancel in pairs; an explosion scatters fragments around an unperturbed COM path.

The standard conservation scenarios (source Q2, Q17, Q22, P22, P26)

Eight standard scenarios; impulse windows; system choice; drills.

Momentum with external forces present: the impulse window (cont.)

Water/rocket windows: variable-mass problems use the dt ledger (CM-16/17) - momentum of the chosen system plus flux terms; never mix the flux ledger with the closed-system shortcut silently.

CHECK / window audit: state 'contact time is short so external impulse is negligible' before using conservation - that sentence is the licence.

(a) Ball bounces off the floor: system = ball alone: floor impulse external, momentum not conserved. System = ball + Earth: conserved (and huge mass makes the recoil unmeasurable). (b) Bullet through block (Q19): system = bullet + block: horizontal p conserved through the passage ok; energy is not (1045 J to heat).

Choosing the system: worked contrasts (cont.)

(c) Two blocks and a spring on ice: system = both + spring: p conserved, and mechanical energy conserved (smooth spring) - the rare case where BOTH hold. (d) Cart under a hopper (OLY-10): system = cart + dropped sand column: horizontal p conserved; the loading force $\dot{m}v$ is internal to the widened system.

CHECK / one-liner test: 'could an external force deliver a noticeable impulse during the event?' If yes and you used conservation, the answer is wrong.

Conservation drills

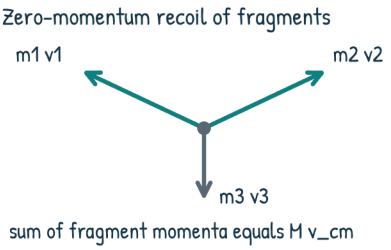
(a) 5 kg cart at 4 m/s picks up 5 kg dropped from rest: $v = 20/10 = 2$ m/s. (b) 3 kg at 6 east collides and sticks to 2 kg at 4 west: $18 - 8 = 10 = 5v$: $v = 2$ m/s east. (c) 0.5 kg ball at 10 hits 1.5 kg at rest, $e = 0.5$: $v_2 = (1+e)m_1u_1/(m_1+m_2) = 1.5(0.5)(10)/2 = 3.75$ m/s; $v_1 = (0.5 - 0.75)10/2 = -1.25$ m/s (rebounds). (d) Two ice skaters 60 and 90 kg push to relative 3 m/s: $60v_1 = 90v_2$, $v_1 + v_2 = 3$: $v_2 = 1.2$, $v_1 = 1.8$ m/s.

Conservation drills (cont.)

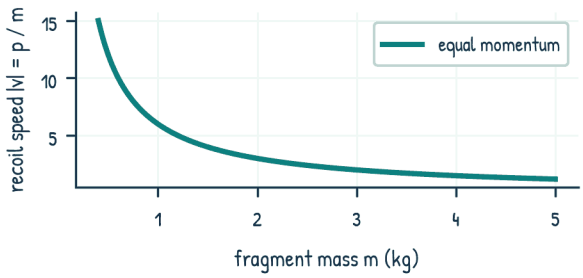
CHECK / drill (c) energy: $K_i = 25$; $K_f = 0.5(0.5)(1.56) + 0.5(1.5)(14.06) = 0.39 + 10.55 = 10.94$ J; with $\mu = 0.375$, $u_{rel} = 10$, loss $= \frac{1}{2}\mu u_{rel}^2(1-e^2) = \frac{1}{2}(0.375)(100)(0.75) = 14.06$ J.

Definitions: ELASTIC = kinetic energy conserved, $e = 1$. PERFECTLY INELASTIC = the bodies stick (maximum KE loss consistent with momentum), $e = 0$. INELASTIC = anything between, $0 < e < 1$. SUPERELASTIC = $e > 1$ with a released spring/explosive - possible, and energy GAINS. The coefficient of RESTITUTION measures speeds along the line of impact:

Zero initial momentum forces fragment momentum vectors to close



Light recoil partner carries most of the kinetic energy p^2/(2m)



REMEMBER / Total momentum is conserved when the net external impulse vanishes, component by component.

Elastic, inelastic, sticking: one number e prices every collision

Restitution definition and scope; line of impact; deformation/reformation impulses; drills.

Elastic, inelastic, sticking: restitution e (cont.)

$e = \frac{v_{sep}}{v_{app}} = \frac{v_2 - v_1}{u_1 - u_2}$ (1-D; see CM-11 for the 2-D projection).

Why e is about speeds not velocities: it is defined on the components along the common normal at contact. For smooth bodies the tangential components are untouched; friction is the only channel across (source trap). Material dependence: e varies with speed and temperature; the exam treats it as a given constant per problem. Energy is a derived quantity here: e and momentum decide the outcome; energy then reports the bill (CM-09).

Elastic, inelastic, sticking: restitution e (cont.)

Sticking = the INELASTIC limit: after contact the bodies share v_{cm} - the only outcome consistent with zero separation speed and momentum conservation. Perfectly inelastic is NOT 'maximum energy destroyed' in one number - it is 'minimum final KE' subject to the momentum ledger; the lost amount is exactly K_{rel} (CM-09).

WHY / the separation/definition choice makes the whole classification one dimensionless number; every exam question about 'what kind of collision' reduces to (i) what is e and (ii) where is the line of impact.

CHECK / limits: $e = 0 \Rightarrow v_2 = v_1 = v_{cm}$ ok; $e = 1 \Rightarrow$ energy conserved ok (verify with the master formulas of CM-07); $e > 1 \Rightarrow$ post-collision KE exceeds pre - flag it as superelastic rather than erroring.

The line of impact (source C1-C2 family)

For SPHERES the line of impact is the line joining centres at the moment of contact. The plane of contact is perpendicular to it. e applies along the line of impact ONLY. For a smooth wall the line of impact is the wall normal. For rough bodies or spinning balls, tangential impulses exist and e no longer closes the system alone - the problem needs friction data.

Angle traps (source C9, corrected): a ball bouncing off a smooth wall leaves at angle θ_r with $\tan\theta_r = \tan\theta_i / e$ from the normal - MORE grazing for $e < 1$; equal angles iff $e = 1$. The mirrored-bounce picture is the elastic-only special case (full correction in CM-22).

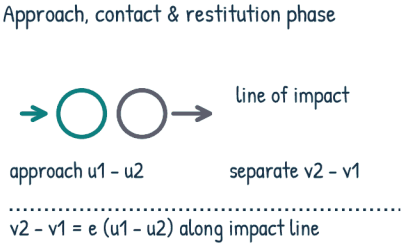
The line of impact (source C1-C2 family) (cont.)

2-D bookkeeping (CM-11): project velocities onto the line of impact, apply momentum + e there, leave tangential components unchanged, recompose. The whole of oblique-collision theory is this three-step recipe.

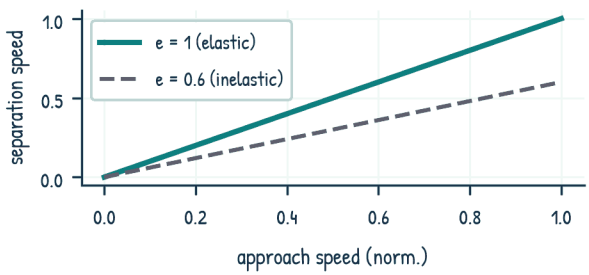
CHECK / draw the contact configuration FIRST: the line of impact is geometry, not choice. A wrong line poisons every later equation.

Every direct impact has two phases. DEFORMATION: the bodies squish to a common velocity v_{cm}; the impulse so far is $J_d = \mu u_{rel}$ (exactly what kills the relative velocity). REFORMATION: elastic recovery pushes them apart with impulse $J_r = e\,\mu u_{rel}$. Total contact impulse on each body along the line: $J = (1+e)\mu u_{rel}$.

Restitution compares post-impact separation to pre-impact approach



Separation speed scales linearly with approach speed for fixed e



REMEMBER / Elastic means e = 1; sticking means e = 0; e measures separation speed over approach speed along the line of impact.

Elastic, inelastic, sticking: one number e prices every collision (cont.)

Restitution definition and scope; line of impact; deformation/reformation impulses; drills.

Impulse of deformation & reformation (Cengage 1.22)

Energy reading: at maximum deformation ALL the relative KE $\frac{1}{2}\mu u_{rel}^2$ is stored; recovery returns the fraction e^2; what never returns is $\frac{1}{2}\mu u_{rel}^2(1-e^2)$. Sticking ($e=0$) has no reformation phase at all; elastic ($e=1$) has $J_r = J_d$ and returns everything.

Impulse of deformation & reformation (Cengage 1.22)

Why $J_d = \mu u_{rel}$ is pure mechanics: at max deformation both bodies share v_{cm} (momentum fixes it whatever the contact law); the impulse that took $(u_1, u_2) \to (v_{cm}, v_{cm})$ is $m_1(v_{cm} - u_1) = -\mu u_{rel}$. Only $J_r = eJ_d$ is the material's business.

CHECK / 2 kg at 5 into 3 kg rest, $e = 0.6$: $J_d = 6$, $J_r = 3.6$, $J = 9.6$ N s; velocities $v_1 = 0.2$, $v_2 = 3.2$; impulse on 3 kg $= 9.6$ ok matches J.

Classification drills and traps

(a) 1 kg at 8 sticks to 3 kg at rest: $v_{cm} = 2$; lost $= K_{rel} = \frac{1}{2}(0.75)(64) = 24$ J; $K_i = 32$, $K_f = 8$: loss 24 ok ($e=0$). (b) Same masses elastic: $v_1 = -4$, $v_2 = 4$; energy conserved ok. (c) $e = 0.7$ version: $v_1 = (1-2.1)8/4 = -2.2$; $v_2 = 1.7(1)(8)/4 = 3.4$; e -check $(3.4 + 2.2)/8 = 0.7$ ok.

WATCH / Trap: 'inelastic means the bodies stick' - only PERFECTLY inelastic does. Trap: using e for energy directly - e is a speed ratio; energy scales as e^2. Trap: applying e to the full speed in 2-D instead of the normal component.

Classification drills and traps (cont.)

CHECK / drill (a) fraction: equal masses lose exactly HALF the KE when sticking ($K_{cm} = K_{rel}$ split) ok - a favourite true/false.

Momentum: $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$.

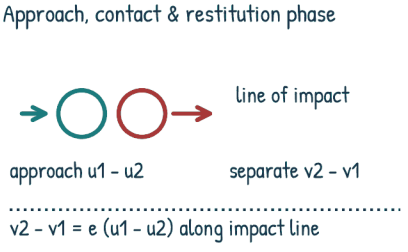
Restitution: $v_2 - v_1 = e(u_1 - u_2)$. Solve the pair for the master formulas:

$$v_1 = \frac{(m_1 - em_2)u_1 + (1+e)m_2u_2}{m_1 + m_2}$$ and

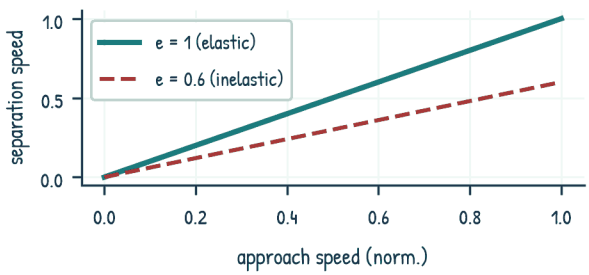
$$v_2 = \frac{(1+e)m_1u_1 + (m_2 - em_1)u_2}{m_1 + m_2}.$$

Valid in 1-D (or along the line of impact in 2-D) for fixed masses; e given; no external impulse.

Restitution compares post-impact separation to pre-impact approach



Separation speed scales linearly with approach speed for fixed e



REMEMBER / Elastic means e = 1; sticking means e = 0; e measures separation speed over approach speed along the line of impact.

The pair that prices every 1-D collision

Master formulas 3.7a/b; three-invariant audit; P32 worked; drills.

The pair that prices every 1-D collision (cont.)

Memory anchors: the shooter's own mass enters its numerator with MINUS e times the target's; the target's velocity is dragged in with $+(1+e)$; the denominator is always the total mass. Limits: $e=0$ gives both velocities $=v_{cm}$ ok; $e=1$, $m_1=m_2$ gives $v_1=u_2$, $v_2=u_1$ (the exchange) ok; $m_2 \gg m_1$ gives $v_1 \approx -u_1+2u_2$ (the wall) ok.

Relative-velocity form: $v_{rel}=-e\,u_{rel}$ with $u_{rel}=u_1-u_2$ - one line, sign-sensitive. Combined with v_{cm} fixed by momentum, this reconstructs both velocities faster than the formulas when $u_2=0$ or masses are equal. The CM frame (CM-15) is the same pair in geometric dress.

The pair that prices every 1-D collision (cont.)

WHY / the pair is complete: two equations, two unknowns. e is the only place material physics enters; everything else is conservation. Any collision answer that violates either equation is simply wrong.

CHECK / three-invariant audit for every computed answer: (1) signed momentum, (2) $e=(v_2-v_1)/(u_1-u_2)$, (3) energy ledger (CM-09). The trio catches sign slips, mis-multiplied factors, and wrong-mass errors.

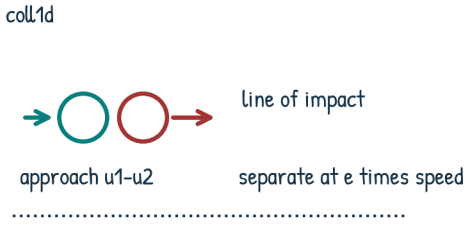
Worked with the invariants: the P32 set (source)

2 kg at 5 m/s hits 3 kg at -2 m/s with $e=0.5$ (source P32). Momentum: $10-6=4$. Master: $v_1=\frac{(2-1.5)(5)+1.5(3)(-2)}{5}=\frac{2.5-9}{5}=-1.3$ m/s; $v_2=\frac{1.5(2)(5)+(3-1)(-2)}{5}=\frac{15-4}{5}=2.2$ m/s. Invariant 1: $2(-1.3)+3(2.2)=-2.6+6.6=4$ ok. Invariant 2:

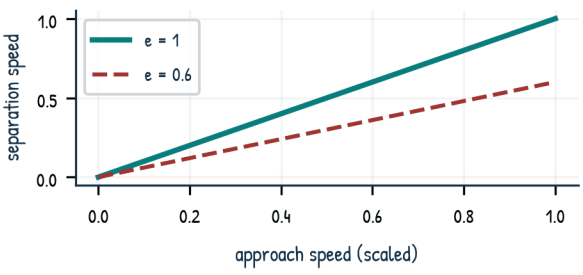
Worked with the invariants: the P32 set (source) (cont.)

$(2.2+1.3)/(5+2)=3.5/7=0.5$ ok. Invariant 3: $K_i=\frac{1}{2}(2)(25)+\frac{1}{2}(3)(4)=31$ J; $K_f=\frac{1}{2}(2)(1.69)+\frac{1}{2}(3)(4.84)=8.95$ J; loss $=22.05$ J, exactly $\frac{1}{2}(1.2)(49)(1-0.25)=22.05$ from the CM-09 formula.

coll1d: labelled physical setup



coll1d: limiting curve / diagnostic



REMEMBER / One pair of equations prices every 1-D collision: momentum conservation plus v2 - v1 = -e (u1 - u2).

The pair that prices every 1-D collision (cont.)

Master formulas 3.7a/b; three-invariant audit; P32 worked; drills.

Worked with the invariants: the P32 set (source) (cont.)

Arithmetic discipline: the factor $(1+e)$ multiplies the WHOLE m_2u_2 term ($1.5\times3\times(-2)=-9$, not -15); one dropped factor here produced the source's Q18 slip family. Write the numerator terms with their factors before multiplying.

CHECK / if any of the three invariants fails, recompute - do not adjust the answer to fit. The three together pin the solution uniquely.

Drill set (each: compute, then audit)

(a) 2 kg at 5 vs 2 kg rest, $e=1$: $v_1=0$, $v_2=5$ (exchange). (b) Same with $e=0.5$: $v_1=1.25$, $v_2=3.75$; e-check $2.5/5=0.5$ ok. (c) 3 kg at 4 vs 3 kg at -4, $e=1$: $v_1=-4$, $v_2=4$ (equal masses swap in the CM: they reverse). (d) 0.5 kg at 12 vs 1.5 kg rest, $e=0.8$: $v_1=(0.5-1.2)12/2=-4.2$; $v_2=1.8(0.5)(12)/2=5.4$; check $(5.4+4.2)/12=0.8$ ok.

Drill set (each: compute, then audit) (cont.)

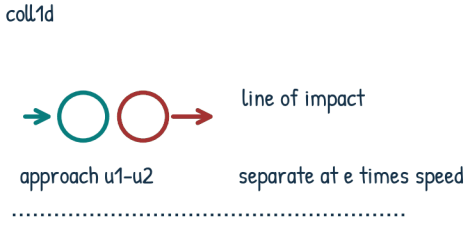
(b2) 0.2 kg at 10 m/s vs 0.3 kg at rest, $e=0.8$: $v_1=(0.2-0.24)10/0.5=-0.8$ m/s (rebounds); $v_2=1.8(0.2)(10)/0.5=7.2$ m/s; audit: $p=2$ both sides; $e=(7.2+0.8)/10=0.8$; $K_{loss}=10-(0.064+7.776)=2.16=\frac{1}{2}(0.12)(100)(0.36)$ - the momentum, restitution and loss checks agree.

CHECK / drill (d) energy: $K_i=36$; $K_f=4.41+21.87=26.28$; loss $=9.72=\frac{1}{2}(0.375)(144)(1-0.64)=9.72$ ok.

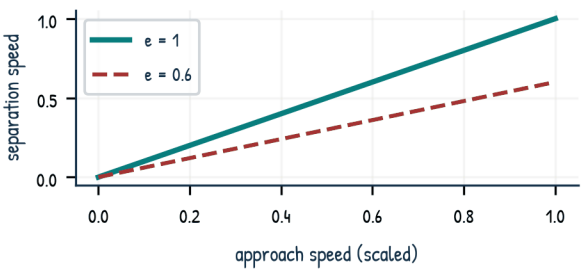
Drill set (each: compute, then audit) (cont.)

Two contacts at once (a wedge between two balls): solve the constraints together or sequentially with clear stages. Variable mass: use the flux ledger (CM-16). Extended bodies with spin: angular impulse needed (rigid-body chapter). Explosions: e is meaningless (no contact) - use momentum + energy release. 2-D: project first (CM-11). The formulas also fail if e is velocity-dependent (real materials at high speed): then the problem must give e at the relevant speed or an energy loss directly. State the constant-e assumption when using them.

coll1d: labelled physical setup



coll1d: limiting curve / diagnostic



REMEMBER / One pair of equations prices every 1-D collision: momentum conservation plus v2 - v1 = -e (u1 - u2).

The pair that prices every 1-D collision (cont.)

Master formulas 3.7a/b; three-invariant audit; P32 worked; drills.

When the master formulas are the wrong tool (cont.)

CHECK / tool-selection audit: does the problem have ONE contact, two fixed masses, and a stated e? If yes use the masters; if not, fall back to first principles.

CASE 1 - equal masses, elastic ($e = 1$): VELOCITIES EXCHANGE: $v_1 = u_2$, $v_2 = u_1$ (source E3). One stops if the other was at rest - in 1-D only; in 2-D the equal-mass elastic result is the 90-degree separation (CM-12). CASE 2 - equal masses, any e: $v_1 = \frac{(1-e)}{2}u_1 + \frac{(1+e)}{2}u_2$ and vice versa - the symmetric split.

Special cases worth recognising on sight (cont.)

CASE 3 - heavy target ($m_2 \gg m_1$): $v_1 \approx (1-e)u_2 - eu_1 +$ corrections; for a wall at rest: $v_1 \approx -eu_1$ (rebounds scaled), $v_2 \approx 0$ (wall barely moves). CASE 4 - light target ($m_1 \gg m_2$): $v_1 \approx u_1$ (plows on), $v_2 \approx (1+e)u_1 - eu_2 \approx 2u_1$ for resting light target and $e = 1$ - the light flies at nearly TWICE the heavy's speed. Q21-set (source):

heavy into light elastic: the light piece's speed is nearly $2u$ and the heavy barely slows (source prints -2 / 1 for its configuration).

Special cases worth recognising on sight (cont.)

CASE 5 - sticking: $v = v_{cm}$ for both. CASE 6 - projectile vs equal target at rest with e: $v_1 = \frac{(1-e)}{2}u$, $v_2 = \frac{(1+e)}{2}u$ - memorise this pair; many drills specialise it. CASE 7 - target moving toward shooter: use the full formulas; the relative-velocity form $v_{rel} = -eu_{rel}$ plus v_{cm} is fastest.

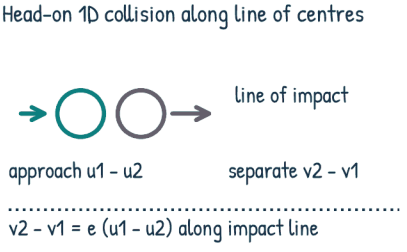
WHY / the cases are the master formulas seen in their natural limits; recognising them converts a 3-minute derive into a 20-second quote - with the derivation still available if the numbers don't match a case.

Q11 worked (the detective problem, source)

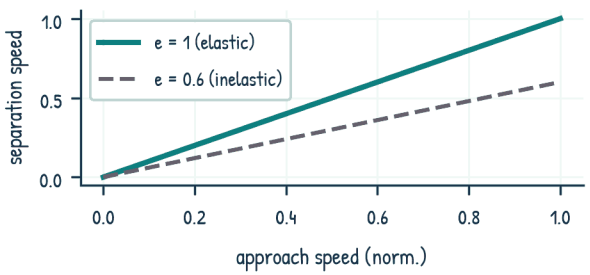
Ball A of mass m at speed u strikes ball B (mass m) at rest; after, A moves at $u/3$ at angle θ_1 (Q11): use the master formulas + the 90-degree theorem (CM-12) to pin the outcome: for equal masses elastic 2-D, $\theta_1 + \theta_2 = 90$ and $v_1 = u\,cos\theta_1$, $v_2 = u\,sin\theta_1$; given one post speed or angle, the rest follows from the momentum triangle.

Method (any Q11-type): (1) check the 90-degree conditions; (2) draw the momentum triangle ($\mathbf{u} = \mathbf{v}_1 + \mathbf{v}_2$); (3) quote the two speeds as components of u; (4) audit KE. If the masses differ or $e < 1$, drop the theorem and run the CM-frame machinery (OLY-02).

Measure all velocities with sign along one chosen impact axis



Elastic limit preserves relative speed while e < 1 damps rebound



REMEMBER / One pair of equations prices every 1-D collision: momentum conservation plus v2 - v1 = -e (u1 - u2).

Special cases worth recognising on sight

Equal-mass exchange, heavy/light limits, sticking; Q11 method; contrast drills.

Q11 worked (the detective problem, source) (cont.)

CHECK / sanity: for any angle the two speeds satisfy $v_1^2 + v_2^2 = u^2$ (elastic equal-mass) - the circle of possible outcomes has radius u.

2 kg at 6 vs 2 kg rest. $e = 1$: $v_1 = 0$, $v_2 = 6$ (exchange). $e = 0.5$: $v_1 = 1.5$, $v_2 = 4.5$; $K_i = 36$ J, $K_f = 2.25 + 20.25 = 22.5$ J, so loss $= 13.5$ J $= \frac{1}{2}(1)(36)(1 - 0.25)$. $e = 0$: both at 3 m/s; loss $= 18 = K_{rel}$ ok.

1 kg at 8 into 3 kg at rest, $e = 1$: $v_1 = (1 - 3)8/4 = -4$ (rebounds); $v_2 = 2(1)(8)/4 = 4$; loss 0 ok. Same with $e = 0$: both at 2 m/s; loss $= 24 = \frac{1}{2}(0.75)(64)$ ok.

Contrast drills: same masses, different e (cont.)

CHECK / the trend table: as e falls from 1 to 0, the shooter's rebound fades into forward drift at v_{cm} and the energy loss rises from 0 to K_{rel} monotonically ok.

(a) equal masses, elastic, one at rest: exchange. (b) heavy hits light elastic at rest: light flies at $\approx 2u$. (c) light hits heavy at rest, $e = 1$: light rebounds at $\approx u$. (d) perfectly inelastic any masses: common v_{cm}. (e) equal masses elastic 2-D: 90 degrees.

Warning: case 1 is 1-D ONLY (source P5's ambiguity; corrected reading in CM-22); case 3's $2u$ needs $e = 1$; case 5's energy loss is K_{rel}, not 'all the energy'.

Recognition drills (60 seconds each) (cont.)

CHECK / self-test: state each case's hypotheses before its conclusion - the hypotheses ARE the exam points.

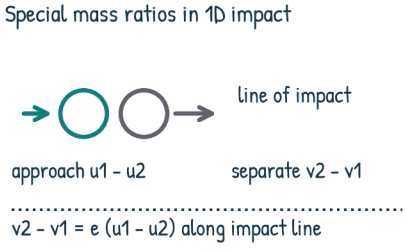
Decompose the KE: $K = \frac{1}{2}Mv_{cm}^2 + \frac{1}{2}\mu v_{rel}^2$ with $\mu = \dfrac{m_1 m_2}{m_1 + m_2}$ (CM-15). Momentum conservation freezes v_{cm}; restitution sets $v_{rel}^{after} = -e\, v_{rel}^{before}$. The collision can only spend the relative part: $K_{loss} = \frac{1}{2}\mu u_{rel}^2(1 - e^2)$ with $u_{rel} = u_1 - u_2$ (source 3.8, P32).

K_loss = 1/2 mu (u1-u2)^2 (1-e^2): the complete energy bill (cont.

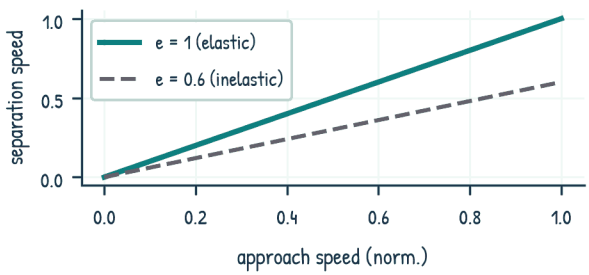
Reading the formula: the STAKE is $K_{rel} = \frac{1}{2}\mu u_{rel}^2$ (the reduced mass and the approach speed); the FRACTION spent is $1 - e^2$ (not $1 - e$). $e = 1$: nothing lost; $e = 0$: the whole K_{rel} gone; superelastic $e > 1$: negative loss = energy released (a spring). The K_{cm} term is untouchable - sticking still leaves $\frac{1}{2}Mv_{cm}^2$ moving.

Worked (P32 numbers): $\mu = 1.2$, $u_{rel} = 7$, $e = 0.5$: $K_{loss} = \frac{1}{2}(1.2)(49)(0.75) = 22.05$ J ok equals $K_i - K_f = 31 - 8.95$ from the master formulas (CM-07). Equal masses sticking: $\mu = m/2$, $u_{rel} = u$: loss $= \frac{1}{4}mu^2 = \frac{1}{2}K_i$ - HALF the energy, the classic true/false (source trap).

Equal elastic masses swap velocities along the line of impact



Restitution e interpolates between sticking (0) and bounce (1)



REMEMBER / Equal masses exchange velocities only in 1-D elastic; heavy barely slows; light bounces at nearly 2u1 - u2.

K_loss = 1/2 mu (u1-u2)^2 (1-e^2): the complete energy bill

K_loss formula and derivation; two-stage ledgers; reduced mass; drills.

K_loss = 1/2 mu (u1-u2)^2 (1-e^2): the complete energy bill (cont.

WHY / the decomposition is exact for two bodies (parallel-axis theorem for KE): $\sum \frac{1}{2} m_i v_i^2 = \frac{1}{2} M v_{cm}^2 + \frac{1}{2} \mu v_{rel}^2$; the cross terms vanish because $\sum m_i \mathbf{v}'_i = 0$. So the collision's budget is literally K_{rel}.

CHECK / E4 (source): 4 kg at 6 into a wall... the perfectly inelastic pair loses K_{rel}: 4 kg at 6 sticks to 1 kg: $\mu = 0.8$, $u_{rel} = 6$: loss $= \frac{1}{2}(0.8)(36) = 14.4$ J; $K_i = 72$, $v = 4.8$, $K_f = 57.6$; loss 14.4 ok. Source E4 prints 183.8 J for its own configuration - same formula.

Energy ledgers in two-stage problems

Ballistic pendulum (CM-13): stage-1 loss $= K_{rel}$ with $\mu \approx m_{bullet}$; the surviving fraction $m/(m+M)$ of the bullet KE then converts to mgh (E5: 800 J in, 3.98 J swing, 796 J heat = 99.5%). Bullet through block (Q19): $K_i = 1250$, $K_f = 204.5$: 1045.5 J heat. Spring problems (CM-14): the collision's loss and the spring's share must be booked separately.

Superelastic bookkeeping: if $e > 1$ or a spring releases during contact, $K_{loss} < 0$; write $K_{gain} = \frac{1}{2} \mu u_{rel}^2 (e^2 - 1)$ and state the source of the energy. Explosions: $\Delta K = +$ chemical/spring release (source E2): no restitution involved.

Energy ledgers in two-stage problems (cont.)

CHECK / double-entry: compute the loss both from $K_i - K_f$ (master formulas) and from $\frac{1}{2} \mu u_{rel}^2 (1 - e^2)$; they must agree to rounding (P32: 22.05 both ways).

$\mu = m_1 m_2/(m_1 + m_2)$: smaller than either mass; for equal masses m: $\mu = m/2$; for $m_2 \gg m_1$: $\mu \approx m_1$ (the light one sets the scale); symmetric in the two bodies. $M = m_1 + m_2$ for the COM term.

The pair (M, μ) is the natural coordinates of two-body motion - the same μ appears in the relative-coordinate equation $\mu a_{rel} = F_{internal}$ and in spring oscillations of two-body systems.

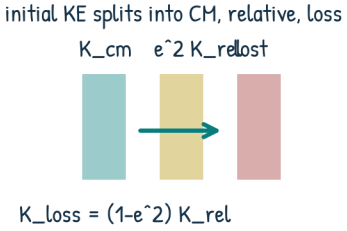
The reduced mass itself (cont.)

Limits: $\mu \to 0$ when one mass vanishes (no collision stake); $\mu \to m/2$ for equal; $1/\mu = 1/m_1 + 1/m_2$ (the harmonic flavour). Drill: 3 kg and 6 kg: $\mu = 2$ kg; 1 kg and 1 kg: 0.5; 1 kg and 1000 kg: 0.999.

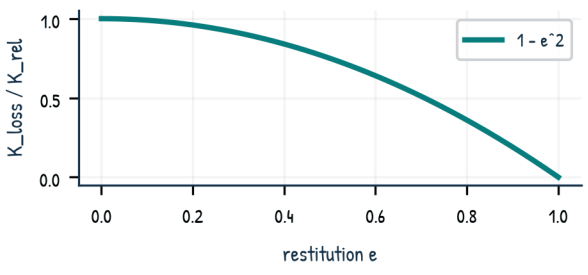
CHECK / dimension: μ is a mass; $\frac{1}{2} \mu v^2$ an energy ok; the loss formula is never negative for $|e| \leq 1$ ok.

(a) 2 kg at 5 vs 3 kg at -2, $e = 0.5$ (P32): 22.05 J. (b) Equal 1 kg masses, $u = 4$, $e = 0.8$: $K_{rel} = 4$, loss $= 4(1 - 0.64) = 1.44$ J. (c) 0.5 kg at 12 vs 1.5 kg rest, $e = 0.8$: $\mu = 0.375$, $u_{rel} = 12$: stake $= 27$, loss $= 27(0.36) = 9.72$ J (matches CM-07 drill (d)).

eloss: labelled physical setup



eloss: limiting curve / diagnostic



REMEMBER / K_loss = 1/2 mu (u1 - u2)^2 (1 - e^2): the reduced mass and approach speed set the stake; e sets the loss.

K_loss = 1/2 mu (u1-u2)^2 (1-e^2): the complete energy bill (cont.)

K_loss formula and derivation; two-stage ledgers; reduced mass; drills.

Loss-formula drills (cont.)

WATCH / Trap: using $1-e$ instead of $1-e^2$; using m_1+m_2 instead of μ; forgetting u_{rel} is SIGNED before squaring (sign dies in the square - but keep it for the e -check).

CHECK / cross-audit every drill against the master-formula difference; the two routes agreeing is the exam's strongest check.

A direct impact has two phases. Phase 1 DEFORMATION: the bodies approach at $u_{rel}=u_1-u_2$, squish, and reach a common velocity v_{cm} at the instant of maximum deformation. The impulse delivered by contact so far is $J_d=\mu u_{rel}$ (exactly what is needed to kill the relative velocity: after it, both bodies move at v_{cm}). Phase 2 REFORMATION: elastic recovery pushes them apart;

Deformation and reformation impulses (cont.)

the reformation impulse is $J_r=e\,\mu u_{rel}$. Total contact impulse on each body: $J=J_d+J_r=(1+e)\mu u_{rel}$, directed along the line of impact.

Why $J_d=\mu u_{rel}$: at maximum deformation both bodies share one velocity v; momentum conservation fixes $v=v_{cm}$ regardless of the contact law. The impulse that took the pair from (u_1,u_2) to (v_{cm},v_{cm}) is $m_1(v_{cm}-u_1)=-\mu u_{rel}$ on body 1 (and opposite on body 2) - pure algebra from momentum, no material assumption.

Deformation and reformation impulses (cont.)

Then the reformation is the material-dependent part: the contact force history after the peak, with impulse scaled by e.

Energy story (this is where the $1-e^2$ comes from): at maximum deformation the ENTIRE relative KE $\frac{1}{2}\mu u_{rel}^2$ has been extracted (stored as deformation energy in the contact). Recovery returns a fraction e^2 of it ($\frac{1}{2}\mu(eu_{rel})^2$ of relative KE after separation). What never comes back: $\frac{1}{2}\mu u_{rel}^2(1-e^2)$ = the permanent loss.

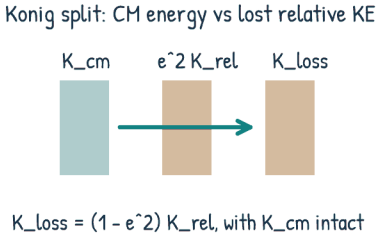
For sticking ($e=0$) there is no phase 2 at all; for elastic ($e=1$) phase 2 returns everything and $J_r=J_d$.

Deformation and reformation impulses (cont.)

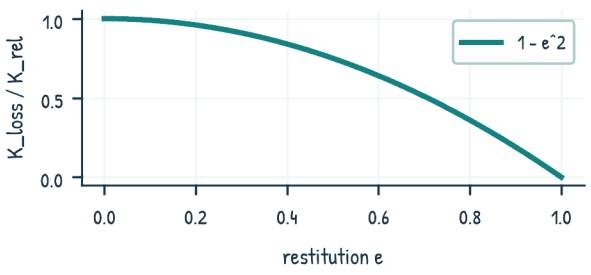
WHY / the decomposition isolates the material law in ONE number (e) while phase 1 is universal mechanics. It also answers 'maximum compression of the spring between the carts' questions: at max deformation the spring holds ALL of the former relative KE; equate $\frac{1}{2}kx_{max}^2=\frac{1}{2}\mu u_{rel}^2$ (CM-14).

CHECK / impulse route reproduces the master formulas: $v_1=u_1-J/m_1$, $v_2=u_2+J/m_2$ with $J=(1+e)\mu u_{rel}$ gives (3.7a-b) term by term. Elastic equal masses: $J=2\mu u=mu$; shooter: $u-mu/m=0$ stops; target: $0+mu/m=u$ takes the speed.

Centre-of-mass kinetic energy K_cm survives every internal impact



Fractional loss of relative kinetic energy equals 1 - e^2



REMEMBER / K_loss = 1/2 mu (u1 - u2)^2 (1 - e^2): the reduced mass and approach speed set the stake; e sets the loss.

Two impulses inside every collision: deformation and reformation

Deformation and reformation impulses J_d = mu u_rel, J_r = e J_d; energy return e^2; peak compression.

Numbers: J_d, J_r and the peak-force language

2 kg at 5 m/s hits 3 kg at rest with $e = 0.6$: $u_{rel} = 5$, $\mu = 1.2$. $J_d = 1.2 \times 5 = 6$ N s; $J_r = 0.6 \times 6 = 3.6$ N s; $J = 9.6$ N s. Check via velocities: $v_1 = (2 - 1.8)5/5 = 0.2$ m/s; $v_2 = 1.6 \times 2 \times 5/5 = 3.2$ m/s. Impulse on 2 kg: $2(0.2 - 5) = -9.6$ N s equals $-J$; on 3 kg: $3(3.2) = 9.6$.

Numbers: J_d, J_r and the peak-force language (cont.)

Peak contact force: at the deformation instant the relative speed is zero and the spring-like contact is maximally compressed - that is where the largest force acts. If a problem quotes an average force over contact time Δt: $\bar{F} = J/\Delta t$; if it quotes a spring constant for the contact: $\frac{1}{2}kx_{max}^2 = \frac{1}{2}\mu u_{rel}^2$ and x_{max} occurs at v_{cm} (same instant).

The deformation instant is the answer to 'when is the spring fully compressed' questions.

Numbers: J_d, J_r and the peak-force language (cont.)

CHECK / deformation-phase energy audit on the numbers above: $K_i = 25$ J; at $v_{cm} = 2$ m/s: $K = \frac{1}{2}(5)(4) = 10$ J; extracted $= 15$ J $= \frac{1}{2}(1.2)(25) = 15$ J $= K_{rel}$. After reformation: $K_f = \frac{1}{2}2(0.04) + \frac{1}{2}3(10.24) = 15.4$ J; returned $= 5.4$ J $= e^2 K_{rel} = 0.36 \times 15$. Lost $= 9.6$ J $= 15(1 - 0.36)$.

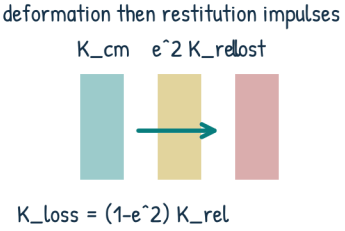
the Cengage floor names these the impulse of deformation (J_d) and impulse of reformation (J_r) in section 1.22; the source chapter does not treat them explicitly - added here as core JEE mechanics.

Where the phase picture is used

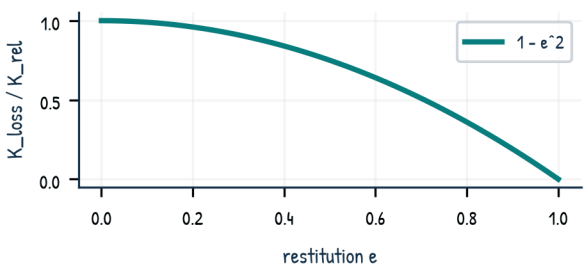
(1) Spring between carts: max compression at the deformation instant when both move at v_{cm}; energy $= K_{rel}$ stored; then release. If the contact is perfectly elastic, the spring returns all of it and the carts separate with $e = 1$ formulas. (2) Squeezing force questions: 'what is the maximum force during contact' - use the contact spring law at x_{max}. (3) Two-stage contact: two carts with a bumper of e then a stop - apply the impulse ledger stage by stage.

(4) Sticking clay: $J = J_d$ only; the whole K_{rel} becomes heat. (5) Real materials: $e < 1$ because the contact hysteresis loop is smaller than the loading loop - $J_r < J_d$; the ratio $J_r/J_d = e$ IS the restitution coefficient in impulse language.

defo: labelled physical setup



defo: limiting curve / diagnostic



REMEMBER / Deformation impulse mu u_app stores the relative KE; a fraction e^2 springs back; (1 + e) mu u_app is the total jolt.

Two impulses inside every collision: deformation and reformation (cont.)

Deformation and reformation impulses J_d = mu u_rel, J_r = e J_d; energy return e^2; peak compression.

Where the phase picture is used (cont.)

WHY / the phase picture survives even when the contact is not a linear spring: $J_d = \mu u_{rel}$ is pure momentum; only $J_r = eJ_d$ is empirical. That is why the same formulas describe clay, steel and rubber.

CHECK / extremes: $e = 0$: $J = \mu u_{rel}$, no return; $e = 1$: $J = 2\mu u_{rel}$, full return. The total impulse is always linear in $(1 + e)$ - a useful graph: J vs e is a straight line.

(a) 4 kg at 3 m/s vs 4 kg at rest, $e = 0.5$: $\mu = 2$, $J_d = 6$, $J_r = 3$, $J = 9$; $v_1 = (4 - 2)3/8 = 0.75$ m/s; $v_2 = 1.5 \times 4 \times 3/8 = 2.25$ m/s; impulse on 4 kg: $4(0.75 - 3) = -9$.

Micro-set: impulse-decomposition drills (cont.)

(b) Sticking pair 2 kg at 6 into 2 kg at rest: $\mu = 1$, $J = J_d = 6$ N s; common speed 3 m/s. (c) Elastic equal-mass exchange: $J = 2\mu u = mu$; the deformation and reformation impulses are equal halves of the contact jolt.

CHECK / drill (a) energy: $K_{rel} = \frac{1}{2}(2)(9) = 9$ J; lost $= 9(1 - 0.25) = 6.75$ J; direct: $K_i = 18$; $K_f = 1.125 + 10.125 = 11.25$; loss $= 6.75$.

Oblique collisions: decompose on the line of impact

For smooth spheres the collision force is purely normal (no friction), so the recipe is fixed: (1) draw the line of impact (the line joining centres at contact) and the plane of contact (perpendicular to it); (2) project both velocities onto the line of impact - these 1-D components obey momentum + e EXACTLY as in CM-07; (3) the components perpendicular to the line of impact (tangential) are UNCHANGED for each body;

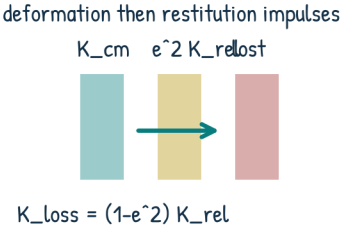
(4) recompose. Momentum is conserved in both directions, but restitution speaks only along the normal.

Oblique collisions: decompose on the line of impact (cont.)

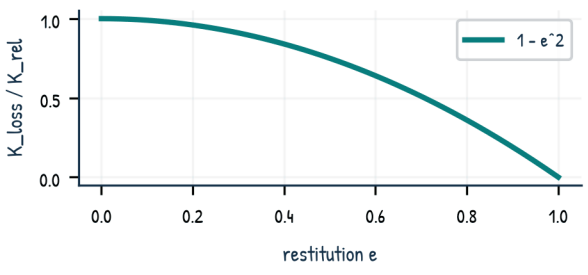
Bookkeeping form: label the line of impact as n and the tangent as t. Then $v_{1n} = \frac{(m_1 - em_2)u_{1n} + (1 + e)m_2u_{2n}}{m_1 + m_2}$, $v_{2n} = \frac{(1 + e)m_1u_{1n} + (m_2 - em_1)u_{2n}}{m_1 + m_2}$, while $v_{1t} = u_{1t}$, $v_{2t} = u_{2t}$. Four scalars in, four out. If the target is at rest, $u_{2n} = u_{2t} = 0$ and the formulas simplify to the zero-target pair of CM-07 along n.

The geometry trap: the line of impact is NOT horizontal unless the geometry says so. For a sphere hitting a fixed inclined plane, the line of impact is the plane normal; for two spheres, it is the line of centres - which depends on the impact parameter (how grazing the hit is). Draw the contact configuration first; every later equation hangs off that line.

defo: labelled physical setup



defo: limiting curve / diagnostic



REMEMBER / Deformation impulse mu u_app stores the relative KE; a fraction e^2 springs back; (1 + e) mu u_app is the total jolt.

Oblique collisions: decompose on the line of impact

Line of impact decomposition; tangential invariance; wall and incline cases; drills.

Oblique collisions: decompose on the line of impact (cont.)

Angles in the final answer are measured from whatever axis the problem states - recompose then project.

WHY / tangential velocities survive because the third-law pair of contact forces acts along the common normal - friction is the only channel into t, and smoothness closes it. This independence is what makes 2-D decomposable into a 1-D problem plus free streaming.

CHECK / recompose-energy sanity for elastic smooth collisions: KE is conserved iff the normal parts conserve it (they do by $e = 1$) and the tangentials are unchanged (they are). If your 2-D elastic answer fails a KE check, the projection angle is wrong.

Worked template: at-rest target, impact parameter set by geomet.

A sphere m_1 moving horizontally at u strikes an identical sphere m_2 at rest; the line of centres at contact makes angle ϕ with the original direction (impact parameter $b = (R_1 + R_2)\sin\phi$ for centre separation $R_1 + R_2$).

Decompose: $u_{1n} = u\cos\phi$, $u_{1t} = u\sin\phi$, target zero.

Elastic equal masses along n: exchange the normal components: $v_{1n} = 0$, $v_{2n} = u\cos\phi$;

tangentials unchanged: $v_{1t} = u\sin\phi$.

Worked template: at-rest target, impact parameter set by geometr.

Recompose: shooter leaves with speed $u\sin\phi$ perpendicular to the line of centres; target with $u\cos\phi$ along the line of centres. The two are PERPENDICULAR - this is the 90-degree theorem of CM-12 in general clothing. For $\phi \to 0$ (head-on) the shooter stops; for $\phi \to 90$ degrees (grazing) nothing happens;

the scattering angle of the shooter is $90 - \phi$ from its original line.

Worked template: at-rest target, impact parameter set by geometr.

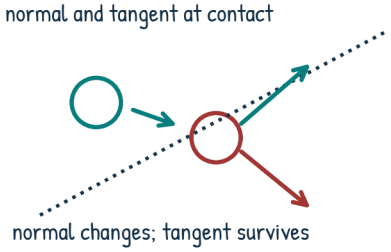
With restitution e and equal masses: $v_{1n} = \frac{(1-e)}{2}u\cos\phi$

from the master: $v_{1n} = \frac{(m-em)u\cos\phi}{2m} = \frac{(1-e)}{2}u\cos\phi$;

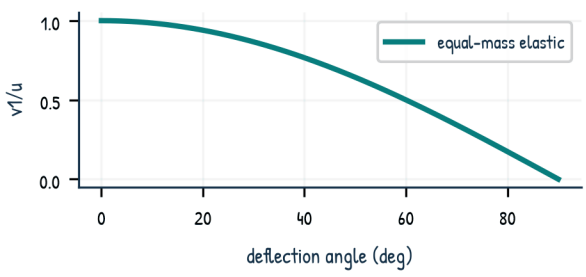
$v_{2n} = \frac{(1+e)}{2}u\cos\phi$. Then $v_{1t} = u\sin\phi$ untouched. The shooter's speed after:

$\sqrt{u^2\sin^2\phi + \frac{(1-e)^2}{4}u^2\cos^2\phi}$ - note only the normal part is taxed by e.

oblique: labelled physical setup



oblique: limiting curve / diagnostic



REMEMBER / Resolve along the line of impact (momentum + e) and perpendicular to it (unchanged for smooth spheres).

Oblique collisions: decompose on the line of impact (cont.)

Line of impact decomposition; tangential invariance; wall and incline cases; drills.

Worked template: at-rest target, impact parameter set by geometr.

CHECK / the $e = 1$, $\phi = 0$ limit gives $v_1 = 0$, $v_2 = u$ (head-on exchange). The $\phi = 90$ degrees limit gives $v_1 = u$, $v_2 = 0$ (no collision). Both limits of the general formula are physically obvious - a good memorisation anchor.

Ball hits a fixed smooth wall at angle θ from the normal with speed v: tangential $= v\sin\theta$ unchanged; normal $= v\cos\theta$ becomes $-ev\cos\theta$ (reversed and scaled). For $e = 1$: reflection with equal angles (specular). For $e < 1$: the rebound is MORE GRAZING: $\tan\theta' = \tan\theta/e$ measured from the normal - the reflected angle exceeds the incident angle. Energy: $K_f/K_i = \sin^2\theta + e^2\cos^2\theta$.

Fixed walls and inclines as 2-D collisions (cont.)

Smooth incline (angle α with horizontal): the line of impact is the incline normal; decompose along the normal and the incline; the normal component flips as e; the along-incline component is untouched. For a ball bouncing off a 30-degree incline elastically, the angle of departure is a pure geometry exercise: reflect the velocity across the normal (mirror rule) - the angle to the incline is preserved.

WHY / the wall is the $m_2 \to \infty$ limit of the two-sphere problem; the wall's velocity is what enters the restitution law ($v_{sep} = e\,v_{app}$ with the wall's motion included, CM-21). The finite-mass formulas degenerate cleanly: the wall never moves.

Fixed walls and inclines as 2-D collisions (cont.)

CHECK / energy check on wall rebound at 45 degrees, $e = 0.8$: $K_f/K_i = 0.5 + 0.64 \times 0.5 = 0.82$; direct: $(v\sin 45)^2 + (0.8v\cos 45)^2 = v^2(0.5 + 0.32) = 0.82v^2$.

(a) 3 kg at 5 m/s along x hits 2 kg at rest; line of impact along x (head-on): back to 1-D, $e = 1$: $v_1 = (3-2)5/5 = 1$ m/s, $v_2 = 2 \cdot 3 \cdot 5/5 = 6$ m/s.

(b) Same but the line of impact is at 60 degrees to x (glancing): normal components: $u_{1n} = 2.5$, $u_{1t} = 4.33$; keep the general formulas along n and add back u_{1t} to v_{1t}.

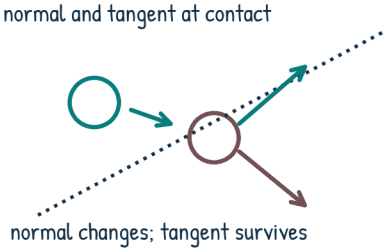
2-D collision drills (cont.)

(c) Ball at 10 m/s at 30 degrees to a smooth floor's normal, $e = 0.75$: rebound normal $= -0.75 \times 8.66 = -6.5$; tangential $= 5$ unchanged; departure angle $= \arctan(5/6.5) = 37.6$ degrees from the normal (more grazing than 30).

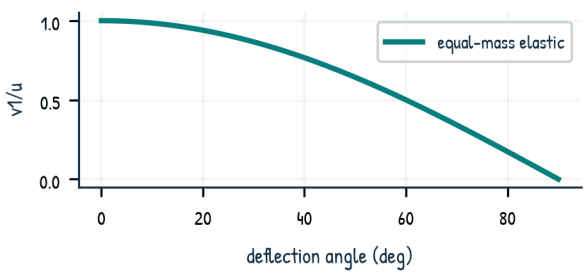
(d) Two 1 kg spheres, one moving at 4 m/s, impact parameter such that $\phi = 45$ degrees, elastic: shooter leaves at 2.83 m/s perpendicular to the centre line; target at 2.83 m/s along it.

CHECK / drill (c) energy: $K_f/K_i = (25 + 42.25)/100 = 0.6725$; formula $\sin^2 30 + e^2\cos^2 30 = 0.25 + 0.5625(0.75) = 0.672$ - agree to rounding.

oblique: labelled physical setup



oblique: limiting curve / diagnostic



REMEMBER / Resolve along the line of impact (momentum + e) and perpendicular to it (unchanged for smooth spheres).

Equal masses, elastic, target at rest: the velocities separate at 90 degrees

Perpendicular-exit theorem via the momentum triangle; scope limits; drills.

Equal masses, elastic, target at rest: the velocities separate at 90 d.

Theorem: if two equal-mass smooth spheres collide elastically and one is initially at rest, their velocity vectors after the collision are perpendicular: $\theta_1 + \theta_2 = 90$ degrees (angles measured from the original direction). Proof in two lines: momentum $m\mathbf{u} = m\mathbf{v}_1 + m\mathbf{v}_2$ gives $\mathbf{u} = \mathbf{v}_1 + \mathbf{v}_2$; elastic KE conservation gives $u^2 = v_1^2 + v_2^2$.

Dot the momentum result with itself: $u^2 = v_1^2 + v_2^2 + 2\mathbf{v}_1 \cdot \mathbf{v}_2$. Comparing: $\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$ - perpendicular. Done. Note the masses cancel in the first equation because they are EQUAL - that is the whole scope condition.

Equal masses, elastic, target at rest: the velocities separate at 90 d.

Speeds from the scatter angle: $v_1 = u\cos\theta_1$, $v_2 = u\sin\theta_1$ where θ_1 is the shooter's deflection (since $\theta_2 = 90 - \theta_1$ and the triangle of momenta is right-angled with hypotenuse mu). Worked (source E6): deflection 30 degrees: $\theta_2 = 60$ degrees; $v_1 = u\cos 30 = 0.866u$; $v_2 = u\sin 30 = 0.5u$.

Momentum-x: $0.866u\cos 30 + 0.5u\cos 60 = 0.75u + 0.25u = u$; momentum-y: $0.866u\sin 30 = 0.433u = 0.5u\sin 60 = 0.433u$.

Equal masses, elastic, target at rest: the velocities separate at 90 d.

Scope and failure modes: (i) unequal masses: NOT 90 degrees - $\theta_1 + \theta_2 < 90$ degrees when $m_1 > m_2$, and the shooter's angle is capped at $\arcsin(m_2/m_1)$ (OLY-02); (ii) $e < 1$: KE drops, the dot-product step fails, the angle is NOT 90 degrees; (iii) rough spheres: tangential impulses change the perpendicular components; (iv) head-on ($\theta_1 = 0$): the shooter stops - the 1-D limit (CM-08).

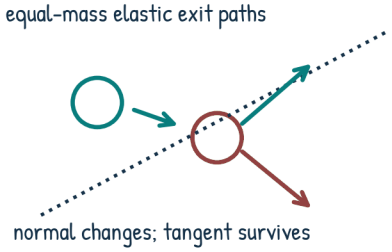
WHY / the dot-product proof works because momentum conservation gives a VECTOR triangle and elasticity gives its SIDE LENGTHS: a right triangle is the only triangle with $a^2 + b^2 = c^2$. Any violation of the theorem's conditions breaks one of those two inputs.

Equal masses, elastic, target at rest: the velocities separate at 90 d.

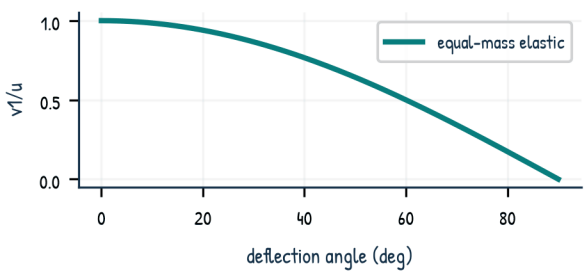
CHECK / source P33: deflection 45 degrees: $\theta_2 = 45$; $v_1 = v_2 = u/\sqrt{2} = 0.707u$; KE: $2 \times \frac{1}{2}m(0.5u^2) = \frac{1}{2}mu^2$ conserved. The equal-angle 45-45 split is the symmetric member of the family.

'Two identical balls, one at rest, elastic, smooth: find the angle between their paths' - 90 degrees, done in one line, no numbers needed. 'One is deflected by 30 degrees: find the other's angle and both speeds' - $\theta_2 = 60$, speeds $u\cos 30$ and $u\sin 30$ (source E6, Q6). 'The collision is NOT smooth (friction): what happens to the 90 degrees?' - it fails:

ninety: labelled physical setup



ninety: limiting curve / diagnostic



REMEMBER / Equal masses, elastic, target at rest: the velocities separate at 90 degrees; head-on is the one-dimensional limit.

Equal masses, elastic, target at rest: the velocities separate at 90 degrees (cont.)

Perpendicular-exit theorem via the momentum triangle; scope limits; drills.

Reading exam questions on the 90-degree rule (cont.)

friction transfers tangential impulse, and (source P33c) the collision may still be elastic ($e = 1$) but directions change; the angle can open or close.

Variant: BOTH balls moving initially - transform to the CM frame first (CM-15): in CM the elastic collision reverses both velocities; transform back. The 90-degree rule is the $u_2 = 0$ special case of that picture. Variant: unequal masses - use the general oblique recipe (CM-11) and let the numbers decide;

the angle between the exits is the sum of the two deflections and no compact closed form is worth memorising: keep the equal-mass theorem and the CM frame instead.

Reading exam questions on the 90-degree rule (cont.)

CHECK / spot-check the momentum triangle: $\mathbf{u} = \mathbf{v}_1 + \mathbf{v}_2$ means the two exit vectors, placed tip to tail, span the original vector exactly. For the 30-60 case: v_1 at 30 degrees $= (0.75, 0.433)u$; v_2 at -60 degrees $= (0.25, -0.433)u$; sum $= (1, 0)u$.

WATCH / Mistake 1: applying it when masses are merely SIMILAR - no: exact equality is the hypothesis. Mistake 2: applying it with $e < 1$ - no. Mistake 3: assuming one ball MUST stop - only in 1-D; in 2-D both move with $u\cos\theta_1$, $u\sin\theta_1$ except at $\theta_1 = 0$ or 90 degrees. Mistake 4:

WATCH / measuring the angle from the wrong axis - the theorem states the angle BETWEEN the two exit velocities, not each vs the incoming line (though $\theta_1 + \theta_2 = 90$ uses those two deflections).

Common mistakes around the theorem (cont.)

WATCH / Mistake 5 (from source P5's ambiguity): 'after the collision the incoming ball stops' - TRUE in 1-D elastic equal-mass (exchange), FALSE in 2-D (deflected). The question 'which happens after the collision' must state the dimension; the safe exam sentence: 'in 1-D the velocities exchange; in 2-D they separate at 90 degrees'.

WHY / the two statements are the same theorem's two limits: the momentum triangle degenerates to a line in 1-D, and the right angle degenerates to the exchange.

CHECK / memorise the one-line proof; regenerating it in 30 seconds beats remembering a formula sheet entry.

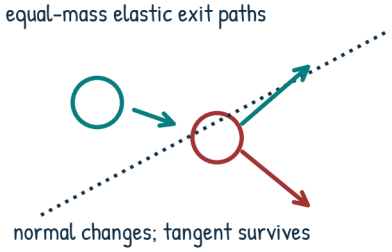
Drills on the 90-degree family

(a) Deflection 37 degrees (3-4-5 triangle!): $\theta_2 = 53$; $v_1 = 0.8u$, $v_2 = 0.6u$. (b) Target twice the mass, elastic: NOT 90 degrees; solve CM-11 style: $v_{1n} = (1-2)/3\, u_n = -u_n/3$ along the normal, tangential unchanged. (c) Equal masses but $e = 0.5$, head-on: $v_1 = u/4$ from master: $(1-0.5)/2\, u = 0.25u$;

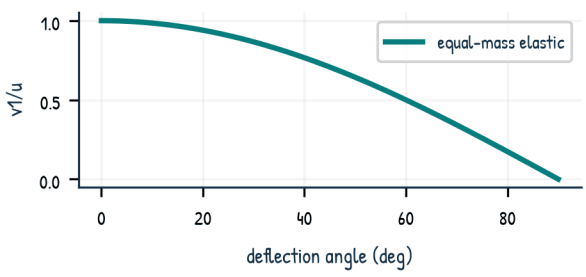
$v_2 = 1.5/2\, u = 0.75u$; angle between them is 0 or 180 (they are collinear) - no 90-degree rule.

CHECK / drill (a): $0.8^2 + 0.6^2 = 1$ - the speed triangle closes on u . Drill (c): e -check $(0.75 - 0.25)/1 = 0.5$.

ninety: labelled physical setup



ninety: limiting curve / diagnostic



REMEMBER / Equal masses, elastic, target at rest: the velocities separate at 90 degrees; head-on is the one-dimensional limit.

Equal masses, elastic, target at rest: the velocities separate at 90 degrees (cont.)

Perpendicular-exit theorem via the momentum triangle; scope limits; drills.

Drills on the 90-degree family (cont.)

in particle physics the same theorem appears in cloud-chamber photographs: equal-mass proton tracks meeting at 90 degrees were used to IDENTIFY the particle as a proton.

Any collision-then-motion problem splits into stages with DIFFERENT conservation laws. Stage 1 (the collision, duration milliseconds): momentum conserved (no external impulse); energy NOT conserved (deformation heat). Stage 2 (the motion afterwards: swing, slide, spring): energy conserved (smooth; only conservative forces); momentum NOT conserved (gravity/tension/spring force is external). The order is fixed:

The two-stage protocol: momentum first, energy second (cont.)

never use energy across the collision, never use momentum across the swing. Write each stage's equation, then chain them through the shared velocity.

Ballistic pendulum (source E5): bullet mass m at v_0 embeds in a hanging block M. Collision: $mv_0 = (m+M)V$, so $V = \dfrac{mv_0}{m+M}$. Swing: $\frac{1}{2}(m+M)V^2 = (m+M)gh$, so $V = \sqrt{2gh}$. Chain: $v_0 = \dfrac{m+M}{m}\sqrt{2gh}$.

The two-stage protocol: momentum first, energy second (cont.)

Worked numbers (source E5): $m = 10$ g, $M = 2$ kg, $v_0 = 400$ m/s: $V = 4/2.01 = 1.99$ m/s; $h = V^2/2g = 3.96/20 = 0.198$ m (about 20 cm).

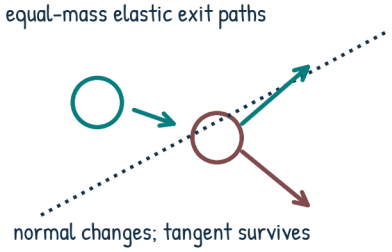
Reverse problems: given h, find v_0: $V = \sqrt{2gh}$ then $v_0 = \frac{m+M}{m}V$. Source Q12: 5 g bullet, 1 kg block, $h = 0.1$ m: $V = \sqrt{2} = 1.414$ m/s; $v_0 = (1.005/0.005)(1.414) = 284$ m/s. Source P29: 10 g, 2 kg, $h = 0.05$ m: $V = 1$ m/s; $v_0 = (2.01/0.01)(1) = 201$ m/s;

The two-stage protocol: momentum first, energy second (cont.)

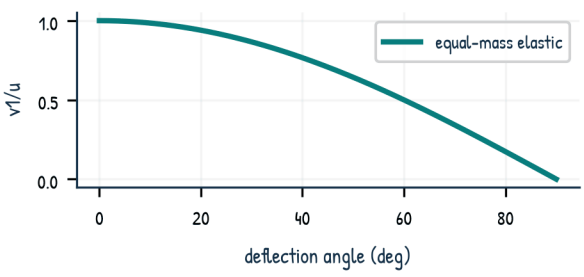
energy lost in the collision: 202 J down to 1.005 J = 99.5% (stage 1 is violently inelastic).

WHY / the collision time is short so gravity's impulse $mg\Delta t$ is negligible - momentum is the right stage-1 law. The swing time is long and the forces are conservative - energy is the right stage-2 law. Mixing them (energy across the embed, momentum across the swing) is the classic examiner trap.

ninety: labelled physical setup



ninety: limiting curve / diagnostic



REMEMBER / Equal masses, elastic, target at rest: the velocities separate at 90 degrees; head-on is the one-dimensional limit.

The two-stage protocol: momentum first, energy second

Ballistic pendulum protocol; bullet-through-block; energy fraction M/(m+M); drills.

The two-stage protocol: momentum first, energy second (cont.)

CHECK / always audit the chain at the shared velocity: if a problem gives v_0 and h, compute V both ways and demand equality. E5: $V = 1.99$ from momentum; $V = \sqrt{2 \times 10 \times 0.198} = 1.99$.

When the bullet EXITS the block (source Q19): 0.01 kg at 500 m/s passes through a 1 kg block at rest and leaves at 200 m/s. One equation fixes the block: $0.01(500) = 0.01(200) + 1(v)$, $v = 3$ m/s. There is no restitution equation to use here (the contact history is complex: entry deformation, exit fracture); momentum alone answers. The energy audit:

Bullet-through-block: two final velocities, one momentum law (con.

$K_i = 1250$ J, $K_f = 200 + 4.5 = 204.5$ J; 1045.5 J went into heat, sound and deformation.

If the block then slides on a rough floor: third stage - work-energy with friction: $\frac{1}{2}Mv^2 = fd$ finds the stopping distance. If it compresses a spring: $\frac{1}{2}Mv^2 = \frac{1}{2}kx_{max}^2$ (CM-14). The protocol generalises: stage 1 momentum; stage 2 energy with whichever conservative/resistive forces apply; sometimes a stage 3.

Bullet-through-block: two final velocities, one momentum law (con.

WATCH / Trap: the ballistic-pendulum formula $v_0 = \frac{m+M}{m}\sqrt{2gh}$ ASSUMES the bullet stays in. If a problem says 'emerges', 'passes through', 'lodges and sticks with extra mass', rewrite stage 1 for that geometry. Same for a bullet that embeds but the block is on a spring (then stage 1 is momentum, stage 2 is spring energy: $\frac{1}{2}(m+M)V^2 = \frac{1}{2}kx^2$).

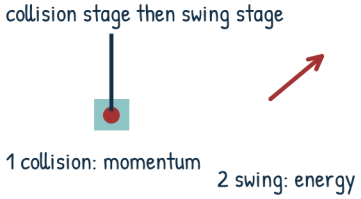
CHECK / Q19 momentum audit with signs: bullet direction positive: $5 = 2 + v$; if the bullet had exited BACKWARD at 200: $5 = -2 + v$, $v = 7$ m/s - the sign convention matters more than the numbers.

Energy fraction in the ballistic pendulum (the examiner favourite)

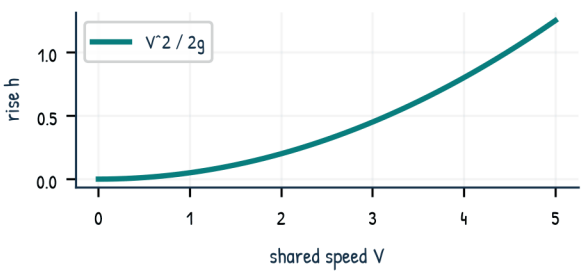
General fraction lost in stage 1: $1 - K_f/K_i = 1 - \frac{m}{m+M} = \frac{M}{m+M}$ (using $V = \frac{m}{m+M}v_0$: $K_f = \frac{1}{2}\frac{m^2 v_0^2}{m+M}$, so $K_f/K_i = \frac{m}{m+M}$). For $M \gg m$: nearly ALL the bullet's KE dies in the block - P29's 99.5%. The retained fraction $m/(m+M)$ keeps the pair moving; the rest is heat and deformation.

Stage 2 then loses nothing (smooth swing) but converts KE to PE. The height reached is $h = \frac{m^2 v_0^2}{2g(m+M)^2}$. For a given h, larger M means larger v_0 measurement sensitivity - the practical ballistic pendulum: measure h, get v_0 with the factor $(m+M)/m$ (a 200x amplification for 10 g in 2 kg).

pendulum: labelled physical setup



pendulum: limiting curve / diagnostic



REMEMBER / Ballistic pendulum: momentum first (the collision), energy second (the swing) - never the other order.

The two-stage protocol: momentum first, energy second (cont.)

Ballistic pendulum protocol; bullet-through-block; energy fraction M/(m+M); drills.

Energy fraction in the ballistic pendulum (the examiner favourite).

WHY / the fraction $M/(m+M)$ is the inelastic energy ledger of CM-09 in disguise: $K_{loss} = K_{rel} = \frac{1}{2}\mu u_{rel}^2$ with $\mu \approx m$, $u_{rel} \approx v_0$; $K_i \approx \frac{1}{2}mv_0^2$; ratio ≈ 1.

CHECK / P29 fraction: $M/(m+M) = 2/2.01 = 99.5$ % exactly the published number. E5: $\frac{1}{2}(0.01)(160000) = 800$ J in; lost $= 796$ J; $K_f = \frac{1}{2}(2.01)(3.96) = 3.98$ J - consistent.

Two-stage drills

(a) 0.02 kg bullet at 300 m/s embeds in 1.98 kg block: $v = 3$ m/s (source P22); if the block hangs on a 0.8 m string, max angle: $h = v^2/2g = 0.45$ m; $\cos\theta = 1 - h/L = 0.4375$, $\theta = 64.1$ degrees. (b) Bullet 5 g at 600 m/s through 1 kg block, exits at 100: $3 = 0.5 + v$, $v = 2.5$ m/s; then the block compresses $k = 500$ N/m spring: $x = v\sqrt{M/k} = 2.5\sqrt{1/500} = 0.112$ m.

(c) 2 kg ball at 3 m/s hits 1 kg at rest with $e = 0.6$; then the 1 kg slides against friction $\mu = 0.2$: $v_2 = 1.6(2)(3)/3 = 3.2$ m/s; friction stop distance: $d = v^2/(2\mu g) = 10.24/4 = 2.56$ m.

Two-stage drills (cont.)

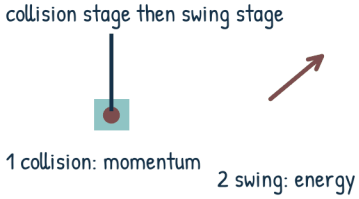
CHECK / drill (a) angle: $h = 0.45$, $L = 0.8$: $1 - 0.45/0.8 = 0.4375$; $\arccos(0.4375) = 64.1$ degrees. Always recompute $h = v^2/2g$ from the stage-1 speed first.

Standard problem (source P35): block m at v_0 slides into block M attached to a spring (k); find the maximum spring compression for (a) perfectly inelastic contact, (b) elastic contact. (a) Sticking: $mv_0 = (m+M)V$, $V = mv_0/(m+M)$; then $\frac{1}{2}(m+M)V^2 = \frac{1}{2}kx^2$:

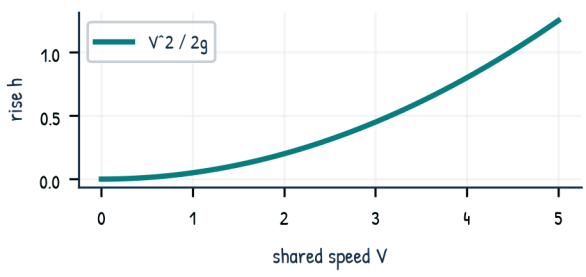
Collision then spring: which compression is larger, sticking or elas.

$x_{in} = \dfrac{mv_0}{\sqrt{k(m+M)}}$. (b) Elastic: target leaves the collision at $v_2 = \dfrac{2m}{m+M}v_0$; then $\frac{1}{2}Mv_2^2 = \frac{1}{2}kx^2$: $x_{el} = \dfrac{2mv_0\sqrt{M}}{(m+M)\sqrt{k}}$.

pendulum: labelled physical setup



pendulum: limiting curve / diagnostic



REMEMBER / Ballistic pendulum: momentum first (the collision), energy second (the swing) - never the other order.

Collision then spring: which compression is larger, sticking or elastic?

Spring compression after collisions; the 3M > m condition; corrected Q18; drills.

Collision then spring: which compression is larger, sticking or elas.

Compare: $\frac{x_{el}}{x_{in}} = 2\sqrt{\frac{M}{m+M}}$. So elastic compresses the spring MORE only when $2\sqrt{M/(m+M)} > 1$, i.e. $4M > m+M$, i.e. $3M > m$. The source's answer 'the elastic collision gives more compression' is CONDITIONAL, not universal:

Collision then spring: which compression is larger, sticking or elas.

if $m > 3M$ the STICKING case compresses more (sticking transfers $\frac{m}{m+M}$ of the KE in one lump; elastic gives the light target $\frac{4mM}{(m+M)^2}$ of the initial KE - compare these fractions directly).
KE fractions check: sticking pair compressing the spring carries $K = \frac{m}{m+M}K_i$. Elastic target carries $K = \frac{4mM}{(m+M)^2}K_i$. Ratio of spring energies $= \frac{4M}{m+M} = (x_{el}/x_{in})^2$ consistent with the amplitude ratio. Equality at $M = m/3$ exactly.

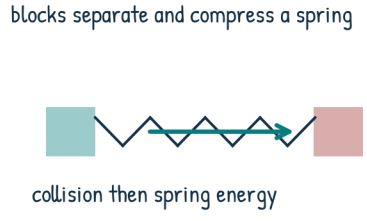
Collision then spring: which compression is larger, sticking or elas.

WHY / the boundary $M = m/3$ is where the two delivery mechanisms cross: elastic fires the target at up to $2v_0$ but leaves the shooter's KE behind; sticking carries everything together at v_{cm} - for a very light target, riding along beats being fired.
CHECK / numbers: $m = 2$, $M = 1$, $k = 500$, $v_0 = 3$: $x_{in} = 6/\sqrt{1500} = 0.155$ m; $x_{el} = 12/67.08 = 0.179$ m; ratio $= 1.155 = 2\sqrt{1/3}$. Since $M = 1 > 2/3 = m/3$, elastic wins here.

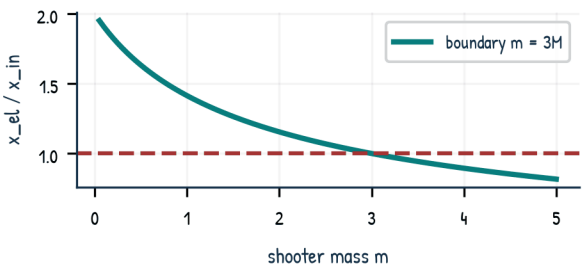
Spring released between two blocks (source Q18, corrected)

2 kg and 3 kg blocks rest on a smooth floor with a compressed spring ($k = 100$ N/m compressed $x = 0.2$ m) between them; released, find the speeds. Momentum: $2v_1 + 3v_2 = 0$, so $v_2 = -2v_1/3$. Energy: $\frac{1}{2}(100)(0.04) = 2$ J $= \frac{1}{2}(2)v_1^2 + \frac{1}{2}(3)v_2^2 = v_1^2 + \frac{3}{2}v_2^2$. Substitute $v_2 = -2v_1/3$: $\frac{3}{2}\cdot\frac{4}{9}v_1^2 = \frac{2}{3}v_1^2$; so $2 = v_1^2 + \frac{2}{3}v_1^2 = \frac{5}{3}v_1^2$: $v_1 = \sqrt{6/5} = 1.095$ m/s, $v_2 = 2(1.095)/3 = 0.730$ m/s.

spring: labelled physical setup



spring: limiting curve / diagnostic



REMEMBER / Post-collision spring compression: elastic beats sticking only when 3M > m; compare with the ratio 2 sqrt(M/(m+M)).

Collision then spring: which compression is larger, sticking or elastic? (cont.)

Spring compression after collisions; the 3M > m condition; corrected Q18; drills.

Spring released between two blocks (source Q18, corrected) (cont.)

Corrections to the source solution (recorded in SOURCE-AUDIT): the source wrote the second KE term as $3 \times 4v_1^2/9 = 4v_1^2/3$ - it dropped the factor $\frac{1}{2}$ in $\frac{1}{2}(3)v_2^2$ - and concluded $v_1^2 = 6/7$, $v_1 = 0.926$, $v_2 = 0.617$.

Those speeds fail the energy audit: $\frac{1}{2}(2)(0.857) + \frac{1}{2}(3)(0.381) = 1.43$ J $\neq 2$ J.

The corrected pair: $\frac{1}{2}(2)(1.2) + \frac{1}{2}(3)(0.533) = 2.0$ J exactly the spring energy.

Spring released between two blocks (source Q18, corrected) (cont.)

WHY / the two-block spring release is just an explosion in disguise: internal forces only, zero total momentum, all spring energy converted to KE of the pair. The split of energy is INVERSELY mass-weighted: lighter block takes more KE ($K_1/K_2 = m_2/m_1$); the momentum split is directly mass-weighted ($v_1/v_2 = -m_2/m_1$).

CHECK / the general check: $U = \frac{1}{2}\mu v_{rel}^2$ with $\mu = 6/5 = 1.2$: $v_{rel} = \sqrt{2U/\mu} = \sqrt{4/1.2} = 1.826$ m/s $= v_2 - v_1 = 0.730 + 1.095 = 1.825$.

Spring-compression variants that appear in exams

(1) Bullet into block attached to spring (P35 with embedding): stage 1 momentum, stage 2 spring energy. (2) Block slides down a ramp into the spring-block: energy through the ramp THEN collision THEN spring - three stages. (3) Two blocks, one spring, one moving (as above). (4) Block hits a SPRING directly and sticks to it - not a collision problem at all:

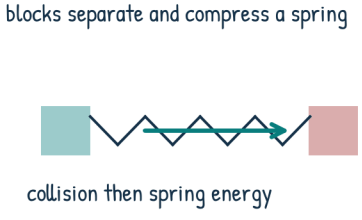
pure SHM with amplitude from energy $\frac{1}{2}mv_0^2 = \frac{1}{2}kA^2$.

Spring-compression variants that appear in exams (cont.)

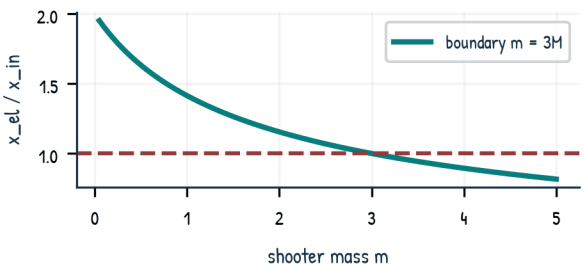
(5) Collision ONTO a spring-mounted target with e : maximum compression is NOT at the end of the contact - during contact the spring compresses further while restitution works; the clean exam route: after separation the target's KE compresses the spring again (detached spring) or, for an ATTACHED spring, the max compression occurs when the two bodies share a velocity (instantaneous rest frame) - solve momentum + energy at that instant.

attached-spring + elastic-contact problems converge to SHM of the relative coordinate with reduced mass μ and spring k ; keep the reduced-mass oscillator as the fallback method.

spring: labelled physical setup



spring: limiting curve / diagnostic



REMEMBER / Post-collision spring compression: elastic beats sticking only when 3M > m; compare with the ratio 2 sqrt(M/(m+M)).

Collision then spring: which compression is larger, sticking or elastic? (cont.)

Spring compression after collisions; the 3M > m condition; corrected Q18; drills.

Spring-compression variants that appear in exams (cont.)

CHECK / variant (4): block 0.5 kg at 4 m/s on $k = 200$: KE $= \frac{1}{2}(0.5)(16) = 4$ J; $A = v\sqrt{m/k} = 4\sqrt{0.0025} = 0.2$ m; $\frac{1}{2}kA^2 = 100(0.04) = 4$ J - consistent.

(a) 1 kg at 5 m/s sticks to 2 kg against $k = 100$: $V = 5/3$; $x = V\sqrt{3/100} = 1.667(0.173) = 0.289$ m. (b) Same elastic: $v_2 = 2(5)/3 = 3.33$; $x = 3.33\sqrt{2/100} = 3.33(0.141) = 0.471$ m;

Post-collision energy drills (cont.)

ratio $= 1.633 = 2\sqrt{2/3}$ ($M = 2 > 1/3$, elastic wins). (c) Q18-corrected pair (1.095, 0.730) rechecked above.

WATCH / Trap: computing max compression from the SHOOTER's KE after an elastic collision (the shooter may rebound and never touch the spring). Trap: forgetting the spring is attached (sticking) vs detached (target flies off after contact). Trap: using $mv_0 = (m+M)V$ when the bullet exits (use Q19 form).

CHECK / drill (a) energy: $K_f = \frac{1}{2}(3)(2.78) = 4.17$ J; $\frac{1}{2}kx^2 = 50(0.0835) = 4.17$.

The CM frame: where collisions become trivial

The CM (centre-of-mass) frame moves with $\mathbf{v}_{cm} = \dfrac{m_1\mathbf{u}_1 + m_2\mathbf{u}_2}{m_1 + m_2}$. In that frame the total momentum is ZERO by construction: $m_1\mathbf{u}'_1 + m_2\mathbf{u}'_2 = 0$, so $\mathbf{u}'_2 = -\frac{m_1}{m_2}\mathbf{u}'_1$ - the two bodies always approach (or recede) along one line, equal and opposite momenta.

The frame is inertial before and after the collision (COM velocity changes only under external impulse) - so Newton's laws hold in it.

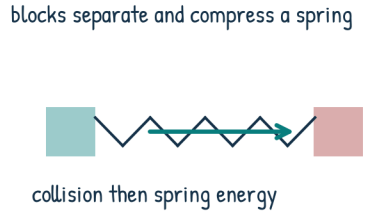
The CM frame: where collisions become trivial (cont.)

Elastic collisions in the CM frame are TRIVIAL: the velocities simply reverse direction (speeds unchanged: $|\mathbf{u}'_i| = |\mathbf{v}'_i|$; directions flipped along the line of impact).

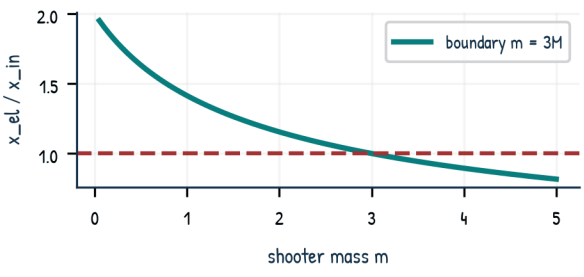
Reason: momentum zero means the momenta reverse (any other outcome would need internal energy change); elasticity keeps the speeds.

In 2-D the reversal happens along the line of impact while the common rotation of both vectors by the scattering angle is free - the magnitude statement is $|v'_i| = |u'_i|$ with direction changed.

spring: labelled physical setup



spring: limiting curve / diagnostic



REMEMBER / Post-collision spring compression: elastic beats sticking only when 3M > m; compare with the ratio 2 sqrt(M/(m+M)).

The CM frame: where collisions become trivial

CM frame definition; velocity reversal for elastic; energy decomposition; circle locus; drills.

The CM frame: where collisions become trivial (cont.)

Energy decomposition: $K = \frac{1}{2}Mv_{cm}^2 + \frac{1}{2}\mu v_{rel}^2$ - the CM-frame KE is $K_{rel} = \frac{1}{2}\mu v_{rel}^2$ where $\mu = m_1 m_2/(m_1+m_2)$ and $v_{rel} = |\mathbf{u}_1 - \mathbf{u}_2|$. First term = the KE of the COM motion (untouchable); second = the KE of relative motion (the collision's play money). For $m_1 = m_2 = m$: $\mu = m/2$, $M = 2m$: $K = mv_{cm}^2 + \frac{m}{4}v_{rel}^2$ - check with $u_2 = 0$: $v_{cm} = u/2$, $v_{rel} = u$: $K = mu^2/4 + mu^2/4 = mu^2/2$.

The CM frame: where collisions become trivial (cont.)

WHY / why the decomposition is exact: $\sum \frac{1}{2}m_i v_i^2 = \frac{1}{2}Mv_{cm}^2 + \frac{1}{2}\mu v_{rel}^2$ for two bodies is the parallel-axis theorem for kinetic energy - expand $\mathbf{v}_i = \mathbf{v}_{cm} + \mathbf{v'}_i$ and the cross terms vanish because $\sum m_i \mathbf{v'}_i = 0$.

CHECK / source P36(a-b): derive the decomposition (done); CM KE is unchanged by the collision because momentum conservation leaves v_{cm} fixed. The lab-frame velocities are just $\mathbf{v}_i = \mathbf{v}_{cm} + \mathbf{v'}_i$ - transform back and you have solved any collision by inspection.

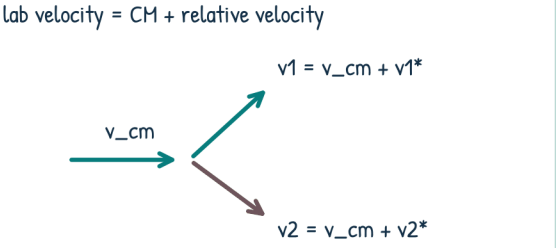
The CM-frame recipe for 2-D scattering

Recipe: (1) compute $\mathbf{v}_{cm}$ and the CM velocities $\mathbf{u'}_i = \mathbf{u}_i - \mathbf{v}_{cm}$; (2) in CM the collision reverses/rotates the velocities (elastic: reverse the relative velocity along the line of impact; keep magnitudes); (3) transform back $\mathbf{v}_i = \mathbf{v}_{cm} + \mathbf{v'}_i$. For a target at rest, $u'_1 = \frac{m_2}{m_1+m_2}u$, $u'_2 = -\frac{m_1}{m_1+m_2}u$ - approach speeds in CM are INVERSELY mass-weighted.

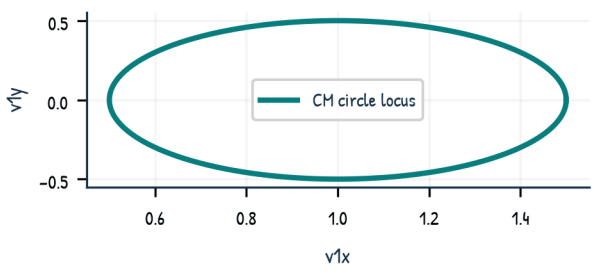
The CM-frame recipe for 2-D scattering (cont.)

The equal-mass 90-degree rule (CM-12) is one line in this language: $\mathbf{v}_{cm} = \mathbf{u}/2$; after reversal $\mathbf{v}_1 = \mathbf{u}/2 + \mathbf{w}$, $\mathbf{v}_2 = \mathbf{u}/2 - \mathbf{w}$ with $|\mathbf{w}| = u/2$ (speeds in CM preserved); $\mathbf{v}_1 \cdot \mathbf{v}_2 = u^2/4 - w^2 = 0$ perpendicular. ANY elastic equal-mass collision with a resting target lands on this circle of radius $u/2$ around $\mathbf{u}/2$ - the locus of possible exit velocities is a circle (useful for maximum/minimum questions!).

cmframe: labelled physical setup



cmframe: limiting curve / diagnostic



REMEMBER / K = 1/2 M v_cm^2 + 1/2 mu v_rel^2; in the CM frame elastic collisions just reverse velocities.

The CM frame: where collisions become trivial (cont.)

CM frame definition; velocity reversal for elastic; energy decomposition; circle locus; drills.

The CM-frame recipe for 2-D scattering (cont.)

Max-deflection questions: the shooter's exit velocity $\mathbf{v}_1 = \mathbf{v}_{cm} + \mathbf{v'}_1$ is the sum of a fixed vector and a vector of fixed length $\frac{m_2}{m_1+m_2}u$ at variable angle. The extreme directions are tangent to that circle - this gives the maximum scattering angle (the OLY-02 formula $\theta_{max} = \arcsin(m_2/m_1)$ for $m_1 > m_2$ in one geometric picture).

CM language for explosions and multi-body problems

CHECK / work E6 in CM: $u = v_0$ along x; $v_{cm} = v_0/2$; CM speeds $v_0/2$ each. Shooter deflected 30 in lab at speed $0.866v_0$: $\mathbf{v}_1 = (0.75, 0.433)v_0$; $\mathbf{v}_1 - \mathbf{v}_{cm} = (0.25, 0.433)v_0$ with length $0.5v_0$ matches CM speed. Target: $\mathbf{v}_2 = \mathbf{v}_{cm} + \mathbf{v'}_2 = (0.5, 0) + (-0.25, -0.433) = (0.25, -0.433)v_0$; length $= 0.5v_0$ - both CM speeds preserved.

Explosion: all fragments' momenta sum to $M\mathbf{v}_{cm}$; in the CM frame they scatter with zero total momentum. For three equal masses: momentum triangle closes (CM-03). For n fragments, the CM frame is where you compute the minimum/maximum possible speed of one fragment: e.g. two fragments fly off with momenta $\mathbf{p}, -\mathbf{p}$ in CM; lab speeds span $|v_{cm} \pm p/m|$.

CM language for explosions and multi-body problems (cont.)

Collisions of more than two bodies: pairwise with e bookkeeping; the CM frame of the PAIR is the natural frame per contact. A ball bouncing between two moving walls: use the wall's frame for each bounce (CM-21), then compose velocities.

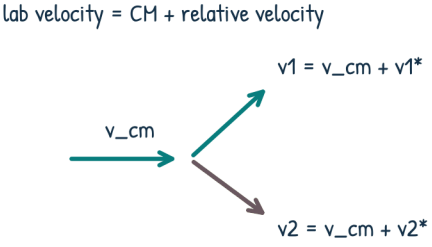
the relativistic generalisation keeps the same structure: $K = \gamma_{cm}Mc^2 + K_{rel}$; the CM frame is still where total momentum vanishes. The non-relativistic decomposition suffices for JEE/Olympiad classical problems.

CM language for explosions and multi-body problems (cont.)

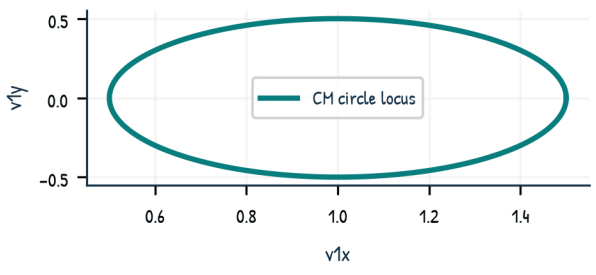
CHECK / invariant audit: v_{cm} is the same before and after any internal rearrangement; K_{cm} likewise. If a computation changes K_{cm}, the momentum bookkeeping is wrong.

(a) 2 kg at 6 m/s hits 1 kg at 2 m/s head-on, elastic: $v_{cm} = (12+2)/3 = 14/3 = 4.667$ m/s; CM speeds: $u'_1 = 6 - 4.667 = 1.333$, $u'_2 = 2 - 4.667 = -2.667$; after reversal: $v'_1 = -1.333$, $v'_2 = 2.667$; lab: $v_1 = 3.333$, $v_2 = 7.333$ m/s. Master-formula check: $v_1 = [(2-1)6 + 2(1)(2)]/3 = 3.333$;

cmframe: labelled physical setup



cmframe: limiting curve / diagnostic



REMEMBER / K = 1/2 M v_cm^2 + 1/2 mu v_rel^2; in the CM frame elastic collisions just reverse velocities.

The CM frame: where collisions become trivial (cont.)

CM frame definition; velocity reversal for elastic; energy decomposition; circle locus; drills.

CM-frame drills (cont.)

$v_2 = [2(2)(6) + (1-2)2]/3 = 7.333$.

(b) Explosion of 4 kg at 10 m/s into 1 kg and 3 kg: $v_{cm} = 10$; CM momenta $\pm p$; lab speeds $10 \pm p/m$. If the 1 kg piece gets $v = 13$ in lab: $p = 1(3) = 3$; 3 kg piece: $10 - 1 = 9$ m/s.

CHECK / drill (a) energies: $K_i = 36 + 2 = 38$; $K_f = \frac{1}{2}(2)(11.11) + \frac{1}{2}(1)(53.78) = 11.11 + 26.89 = 38$ - the CM frame never lies.

Picking mass up and throwing it down: F = m_dot v

A conveyor belt runs at speed v; sand falls on it at rate $\dot{m}$ (kg/s) with zero horizontal velocity. To carry the sand, the belt must deliver to each kilogram the momentum v: force $F = \dot{m}v$. Worked (source E7): $\dot{m} = 5$ kg/s, $v = 2$ m/s: $F = 10$ N. Power delivered by the motor: $P = Fv = \dot{m}v^2 = 20$ W. But the sand's KE grows at only $\frac{1}{2}\dot{m}v^2 = 10$ W.

The missing 10 W is heat - friction between sand and belt scatters the KE as the grains slide to rest in the belt frame.

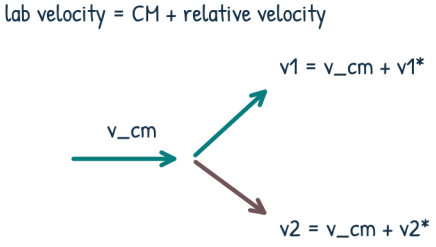
Picking mass up and throwing it down: F = m_dot v (cont.)

The factor-2 is universal for inelastic pickup from rest: belt power $= \dot{m}v^2$, KE rate $= \frac{1}{2}\dot{m}v^2$, heat rate $= \frac{1}{2}\dot{m}v^2$. Examiners love the 'where does the other half go' question (source E7/P34): answer: sliding friction heat at the pickup (and if the sand is thrown OFF the belt at the far end, that KE is carried away with the sand - different audit!).

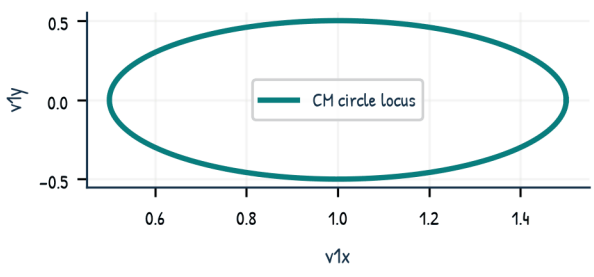
Picking mass up and throwing it down: F = m_dot v (cont.)

Cart collecting sand (OLY-10): a cart of mass M rolling at v on a smooth floor collects rain/sand falling vertically at $\dot{m}$: horizontal momentum is conserved (vertical rain has none), so $Mv = (M + \dot{m}t)v(t)$: $v(t) = \dfrac{Mv_0}{M + \dot{m}t}$ - the cart slows. A cart that LEAKS sand vertically:

cmframe: labelled physical setup



cmframe: limiting curve / diagnostic



REMEMBER / K = 1/2 M v_cm^2 + 1/2 mu v_rel^2; in the CM frame elastic collisions just reverse velocities.

Picking mass up and throwing it down: F = m_dot v

F = m_dot v; belt power split; cart pickup and leak; sign discipline; drills.

Picking mass up and throwing it down: F = m_dot v (cont.)

v constant (the dropped sand takes its own momentum); a cart REJECTED from a hopper at speed u rel needs thrust $\dot{m}(u-v)$ - sign bookkeeping always via momentum of (cart + ejected mass) per dt.

WHY / the pickup force is momentum flux: mass $\dot{m}$ per second crosses from speed 0 to speed v; the momentum source is the belt, paid as $\dot{m}v$. Same law as the rocket (CM-17) and the falling chain (OLY-01) with different geometries.

CHECK / P34(a): $\dot{m}=3$ kg/s at $v=4$ m/s: $F=12$ N, $P=48$ W, KE rate $=24$ W, heat $=24$ W - exactly the factor-2 split.

Sand thrown off: the full belt audit

If sand lands at rest and leaves the belt at speed v (belt length L, speed v): power in $\dot{m}v^2$; KE out $\frac{1}{2}\dot{m}v^2$ carried away; heat $\frac{1}{2}\dot{m}v^2$. If instead the belt STOPS the sand (sand lands moving at v and the belt brings it to rest): force $=\dot{m}v$ still, but now the KE $\frac{1}{2}\dot{m}v^2$ is destroyed entirely - all heat.

The audit table depends on in/out speeds; the momentum flux law $F=\dot{m}(v_{out}-v_{in})$ never changes.

Sand thrown off: the full belt audit (cont.)

Hopper onto a cart: $F=\dot{m}u$ if the cart must accelerate the sand from rest to its own speed u; the cart's engine force F at speed u delivers $P=\dot{m}u^2$ of which half is KE rate - only if the sand starts at rest in the lab. If the sand is thrown forward at w relative to the cart: thrust bookkeeping via the rocket sign discipline (CM-17): the safest exam route is to write momentum of the whole system at t and $t+dt$ and expand.

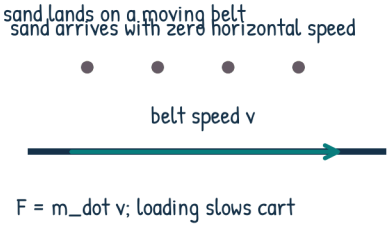
CHECK / cart collecting rain at 0.1 kg/s from 5 m/s at $t=0$, $M=20$ kg: at $t=100$ s: $v=100/30=3.33$ m/s; KE: 250 J to $\frac{1}{2}(30)(11.1)=166.7$ J; the 83.3 J lost is the pickup heat, matching the differential identity $\frac{1}{2}\dot{m}v^2=-\frac{d}{dt}KE$.

Sand thrown off: the full belt audit (cont.)

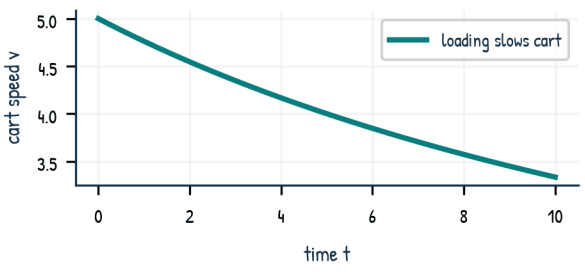
the sand-on-belt P = 2 x KE-rate result reappears in fluid jets (OLY-07: F = rho A v^2 on a plate) and the falling chain (OLY-01: flux 2 lambda g x) - one momentum-flux law, many costumes.

The universal recipe: system = (body + the mass being exchanged) over dt; write $p(t)$ and $p(t+dt)$; keep every term to FIRST order in dt; drop products like $dm\,dv$. The force on the body: $F_{on\,body}=\dot{m}_{in}u_{in}-\dot{m}_{out}u_{out}$ (momentum brought in minus carried out) plus external forces. For the rocket: $m\,dv=-v_e\,dm$ with $dm<0$ (mass lost) - signs set by the exhaust velocity RELATIVE to the body.

belt: labelled physical setup



belt: limiting curve / diagnostic



REMEMBER / Picking mass up or throwing it down needs force m_dot v; belt power splits half KE, half heat.

Picking mass up and throwing it down: F = m_dot v (cont.)

F = m_dot v; belt power split; cart pickup and leak; sign discipline; drills.

Variable-mass sign discipline (cont.)

WATCH / Traps: (1) using the sand's speed instead of the RELATIVE speed for rockets (CM-17); (2) forgetting the sand carries momentum away when it leaves a cart (that IS the retarding force on leaky carts); (3) assuming energy conservation anywhere in these problems - variable-mass pickup is dissipative by nature.

CHECK / leaky cart: M moving at v leaks at $\dot{m}$ vertically down. Horizontal momentum of cart: the leaked grains keep horizontal speed v; the leak takes $\dot{m}v$ per second but the cart's remaining momentum per mass stays: $\dot{v} = 0$ - v CONSTANT (the classic result: leaking does not slow the cart; loading does).

Drills: flux and belt

(a) Belt 1.5 m/s, sand 2 kg/s: $F = 3$ N, $P = 4.5$ W, KE rate 2.25 W. (b) Rain 0.02 kg/s into a 100 kg cart at 2 m/s: $F = 0.04$ N retarding at first; after 100 s the cart holds 2 kg of water and moves $200/102 = 1.96$ m/s. (c) Water jet 3 kg/s at 8 m/s onto a wall (stops): $F = 24$ N (OLY-07).

(d) Belt problem with friction coefficient given: if the question gives μ between sand and belt, the static friction force on the sand is $\dot{m}v$ total; belt tension difference equals it - solve as a friction-engineering problem, not a collision.

Drills: flux and belt (cont.)

CHECK / drill (b): momentum 200 kg m/s conserved; total mass 102; $v = 1.9608$; KE: initial 200 J; final $\frac{1}{2}(102)(3.845) = 196.1$ J; lost 3.9 J = pickup heat.

Derivation in the lab frame (source 3.9-3.10): rocket mass m, speed v; in time dt it ejects mass $-dm > 0$ at exhaust speed v_e RELATIVE to the rocket (backward), so exhaust lab speed $= v - v_e$. Momentum conservation for rocket + exhaust (no external impulse): $mv = (m + dm)(v + dv) + (-dm)(v - v_e)$.

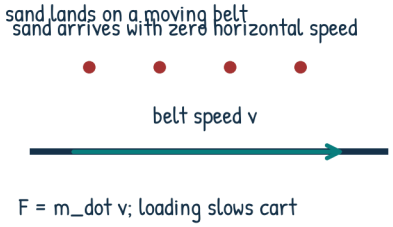
Expand:

Delta v = v_e ln(m0/m_f): the rocket equation (cont.)

$mv = mv + m\,dv + v\,dm + dm\,dv - v\,dm + v_e\,dm$; cancel and drop the second-order $dm\,dv$: $m\,dv + v_e\,dm = 0$, so $dv = -v_e\,dm/m$. Integrate from m_0 to m_f: $\Delta v = v_e \ln \dfrac{m_0}{m_f}$.

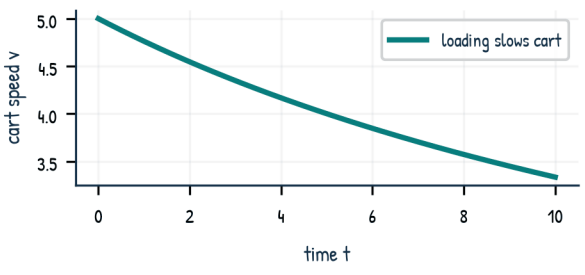
Reading the formula: Δv depends on the exhaust speed and the MASS RATIO only - not on the burn rate, not on the time profile. Gravity-free, drag-free, constant v_e - those are the assumptions (state them). With gravity acting upward during vertical climb: $dv = -v_e dm/m - g\,dt$ and the integral gains $-gt_{burn}$. Thrust:

belt: labelled physical setup



F = m_dot v; loading slows cart

belt: limiting curve / diagnostic



REMEMBER / Picking mass up or throwing it down needs force m_dot v; belt power splits half KE, half heat.

Delta v = v_e ln(m0/m_f): the rocket equation

Rocket equation derivation; thrust; assumptions; multi-burn and drills.

Delta v = v_e ln(m0/m_f): the rocket equation (cont.)

$F = \dot{m} v_e$ with $\dot{m} = |dm/dt|$ (source Q25: $\dot{m} = 2$ kg/s at 3000 m/s: $F = 6000$ N).

Numbers (source E10): $v_e = 3$ km/s, $m_0/m_f = 10$: $\Delta v = 3 \ln 10 = 3(2.303) = 6.91$ km/s - orbit-class.

Source Q8: need $\Delta v = 4$ km/s with $v_e = 2$ km/s: $m_0/m_f = e^2 = 7.39$. Source P24: $v_e = 2500$, $\Delta v = 5000$: $m_0/m_f = e^2 = 7.39$. Note the answer to 'how much fuel' is EXPONENTIAL in the mission Δv - the tyranny of the rocket equation.

WHY / why no air is needed: the rocket pushes on its own exhaust, not on the atmosphere. Source P16: rockets work in vacuum. The momentum ledger is closed: every bit of forward momentum of the rocket is paid by backward exhaust momentum.

Delta v = v_e ln(m0/m_f): the rocket equation (cont.)

CHECK / the limit $m_0/m_f = e$ gives $\Delta v = v_e$ (about 63% of the launch mass is propellant); $m_f \to m_0$ (no fuel) gives $\Delta v = 0$; doubling m_0/m_f adds $v_e \ln 2 \approx 0.69 v_e$ - logarithmic, not linear.

Thrust $F = \dot{m} v_e$ is independent of rocket speed (in the rocket's frame the exhaust leaves at v_e regardless). Power delivered to the ROCKET: $P = Fv = \dot{m} v_e v$ (speed-dependent!); the exhaust KE rate is $\frac{1}{2}\dot{m} v_e^2$ - both are frame-dependent bookkeeping; the invariant statement is momentum flux.

In the instantaneous rest frame of the rocket, the exhaust carries away KE $\frac{1}{2}\dot{m} v_e^2$ in that frame.

Thrust and power; the burn-rate freedom (cont.)

Multi-burn problems: apply $\Delta v = v_e \ln(m_{start}/m_{end})$ per stage/burn and ADD the Δv s (they are speed increments along the path; for collinear burns). Two-stage rockets (OLY-06): the real gain is dropping dead structure; with ZERO structural mass any split gives the same $\Delta v = v_e \ln(M/M_p)$.

with gravity: $\Delta v = v_e \ln(m_0/m_f) - g t_b$ for a vertical burn of duration t_b; the gravity loss is why real launches turn downrange quickly.

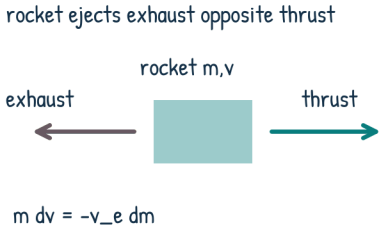
CHECK / Q25 audit: thrust 6000 N on 1000 kg initial gives $a = 6$ m/s^2 at burn start - half a g; the mass falls as fuel burns so acceleration grows. Sign check: $dm < 0$; $-v_e dm/m > 0$ speeds up.

The variable-mass ledger for exams

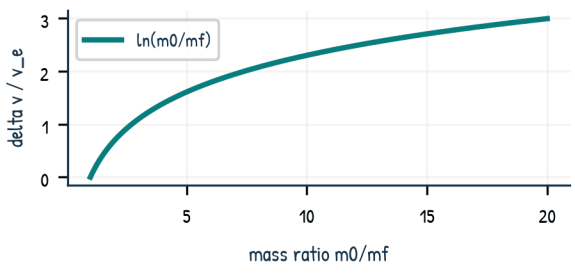
Choose system: rocket + exhaust (closed). Write momentum at t and $t + dt$. Keep first order. Integrate. Assumptions to state: (1) v_e constant relative to the rocket; (2) exhaust collinear; (3) no external impulse (or add g term); (4) all mass ejected is propellant. The same ledger solves sprinklers (water leaving a cart), sand hoppers (CM-16) and jetpacks.

Sprinkler-on-a-cart: cart mass M (with water), ejects at $\dot{m}$ at v_e rel: same equation $Mdv = v_e(-dM)$ if no external force; the cart accelerates like a rocket until the water runs out. Fire-extinguisher recoil: treat the spray as exhaust with v_e = spray speed relative to the cylinder.

rocket: labelled physical setup



rocket: limiting curve / diagnostic



REMEMBER / Rocket: delta v = v_e ln(m0/m_f); thrust = m_dot v_e; the exhaust momentum is the point, not the pushed air.

Delta v = v_e ln(m0/m_f): the rocket equation (cont.)

Rocket equation derivation; thrust; assumptions; multi-burn and drills.

The variable-mass ledger for exams (cont.)

WHY / the rocket equation is just $F = dp/dt$ for a system with flowing mass: $p = mv$ differentiates to $m\,dv + v\,dm$; the exhaust takes the $v\,dm$ term away at speed $v - v_e$ so the leftover is $m\,dv + v_e\,dm = 0$.

CHECK / extinction case: $m_f = m_0/2$: $\Delta v = v_e \ln 2 = 0.69 v_e$; half the mass as fuel gives only 0.69 exhaust-speeds - the classic 'rockets are inefficient' table entry.

Rocket drills

(a) $v_e = 2$ km/s, fuel 80% of mass: $m_0/m_f = 5$: $\Delta v = 2 \ln 5 = 3.22$ km/s. (b) Need 9 km/s with $v_e = 3$: $m_0/m_f = e^3 = 20.1$ - 95% fuel. (c) Thrust 20 kN at $\dot{m} = 50$ kg/s: $v_e = 400$ m/s. (d) A 5 kg cart with a 2 kg spring-loaded mass shot at 6 m/s rel: cart + remaining 3 kg at v; ejected 2 kg at $v - 6$; momentum: $0 = 3v + 2(v - 6) = 5v - 12$; $v = 2.4$ m/s.

CHECK / drill (d) energy: cart KE $= \frac{1}{2}(3)(5.76) = 8.64$ J; ejecta KE $= \frac{1}{2}(2)(3.6^2) = 12.96$ J; total 21.6 J from the spring. Momentum $= 7.2 - 7.2 = 0$.

Chains and jets: momentum flux is m_dot v on top of the carried w.

Chain lifted at constant speed v from a pile on the ground (source P34c, Q13 family): at the moment a length x hangs clear, the force at the top must (1) support the hanging weight $\lambda x g$ and (2) supply the momentum flux to jerk each stationary link up to speed v: $F = \lambda x g + \lambda v^2$ with $\lambda = M/L$. P34(c) with $M = 2$ kg, $v = 3$ m/s, $g = 10$: $F = (2/L)(10x + 9)$ N.

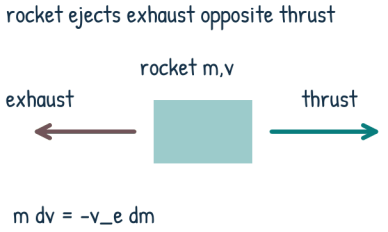
The source's Q13 (mass 3 kg at 2 m/s) is under-specified without L or x - the general formula is the answer (recorded in SOURCE-AUDIT).

Chains and jets: momentum flux is m_dot v on top of the carried w.

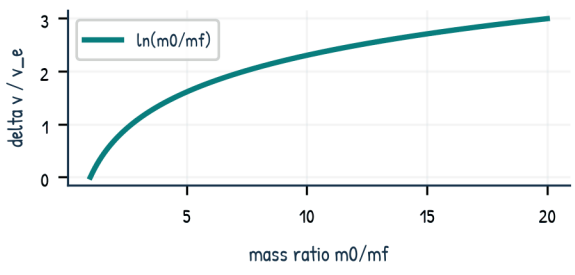
Energetics of the lift (the exam's 'where does the work go'): power delivered $P = Fv = \lambda x g v + \lambda v^3$. Rate of PE growth $= \frac{d}{dt}(\frac{1}{2}\lambda x^2 g) = \lambda x g v$ matches the weight term. Rate of KE growth $= \frac{d}{dt}(\frac{1}{2}\lambda x v^2) = \frac{1}{2}\lambda v^3$. So HALF the flux work λv^3 becomes KE and HALF is dissipated jerking the links from rest to v at the pickup (inelastic).

The pile itself does no work: pickup happens at zero displacement - the chain links at rest are accelerated by tension acting over vanishing displacement; the energy audit closes only with the pickup heat included.

rocket: labelled physical setup



rocket: limiting curve / diagnostic



REMEMBER / Rocket: delta v = v_e ln(m0/m_f); thrust = m_dot v_e; the exhaust momentum is the point, not the pushed air.

Chains and jets: momentum flux is m_dot v on top of the carried weight

Flux law F = m_dot v; chain lift with pickup heat; jets; drills and classification.

Chains and jets: momentum flux is m_dot v on top of the carried w.

Water jet on a fixed plate (OLY-07, source OL9): jet speed v, area A, density ρ: mass rate $\dot{m} = \rho A v$; the horizontal momentum is destroyed (splatters sideways): $F = \dot{m} v = \rho A v^2$. At angle θ to the plate surface: only the normal component $v \sin\theta$ dies: $F_n = \rho A v^2 \sin\theta$. Moving plate at u (same direction as jet): relative speed $v - u$:

$F = \rho A (v - u)^2$ - the flux law always uses relative speed (details in OLY-07).

Chains and jets: momentum flux is m_dot v on top of the carried w.

WHY / one law: $F = \frac{dp}{dt} = \dot{m} \Delta v$. When mass arrives at rest and is carried at v: $\Delta v = v$; when a jet dies on a plate: $\Delta v = v$; when a chain is jerked: $\Delta v = v$; when a rocket ejects: $\Delta v = v_e$. Every flux problem is this multiplication with a diagram.

CHECK / jet numbers: $A = 5$ cm^2, $v = 20$ m/s, $\rho = 1000$: $\dot{m} = 10$ kg/s, $F = 200$ N - about 20 kg-weight, a fire-hose kick.

Chains and jets: momentum flux is m_dot v on top of the carried w.

A chain segment lands and stops: the flux force stopping it is λv^2 where v is the landing speed. If the segment falls a distance x: $v^2 = 2gx$, flux force $= 2\lambda g x$; plus the landed weight $\lambda x g$ gives a total $3\lambda x g$ - the classic result: the scale reads THREE times the landed weight at every instant, ending at $3mg$ when the last link lands.

The chain-lift flux λv^2 is the same term seen from above (OLY-01 does the full derivation and the variant geometry).

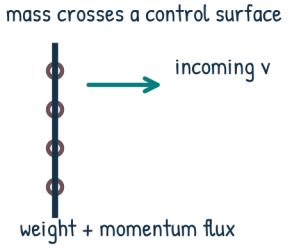
Chain landing on a scale (preview of OLY-01) (cont.)

Exam phrasings to recognise: 'a uniform chain of linear density λ falls onto a scale' vs 'is lifted from a pile at constant speed' vs 'hangs over a smooth peg' (OLY-05) - three geometries, one flux law plus one weight law.

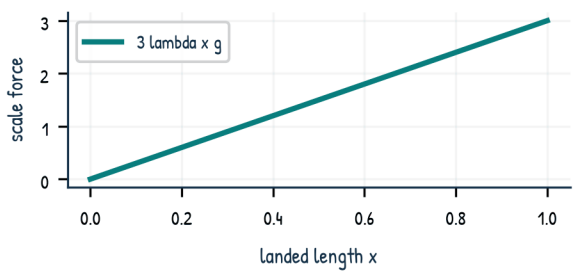
the inelastic pickup/landing heat is exactly $\frac{1}{2}\dot{m}v^2$ at every link - the same half-half split as the belt (CM-16).

CHECK / dimension: $\lambda v^2 =$ kg/m $\times$ m^2/s^2 = kg m/s^2 = N; $\lambda x g$ = kg/m $\times$ m $\times$ m/s^2 = N. Both terms are forces, adding is legitimate.

flux: labelled physical setup



flux: limiting curve / diagnostic



REMEMBER / Momentum flux: F = m_dot v = lambda v^2 on top of the carried weight; jets push rho A v^2.

Chains and jets: momentum flux is m_dot v on top of the carried weight (cont.)

Flux law F = m_dot v; chain lift with pickup heat; jets; drills and classification.

Chain-lift and jet drills

(a) Chain 4 kg, 2 m long, lifted at constant 3 m/s: when 1.2 m is clear: $F = 2(1.2)(10) + 2(9) = 24 + 18 = 42$ N. (b) Jet $\rho = 1000$ kg/m^3, $A = 3$ cm^2, $v = 15$ m/s on a wall: $F = \rho A v^2 = 1000(3 \times 10^{-4})(225) = 67.5$ N. (c) Jet at 60 degrees to a plate surface: $F_n = \rho A v^2 \sin 60 = 0.866\,\rho A v^2$; at 30 degrees to the surface: $0.5\,\rho A v^2$ - the angle is measured to the SURFACE so that the normal component is $v \sin\theta$.

Chain-lift and jet drills (cont.)

(d) Belt at 4 m/s with 6 kg/s sand from rest: $F = 24$ N; $P = 96$ W; KE rate 48 W; heat 48 W (CM-16 again). (e) Chain falling onto a scale: reading $= 3\lambda x g$ at landed length x (OLY-01).

CHECK / drill (a): $\lambda = 2$; $F = \lambda x g + \lambda v^2 = 24 + 18 = 42$ N. Drill (b): $\dot{m} = \rho A v = 1000(3 \times 10^{-4})(15) = 4.5$ kg/s; $F = \dot{m} v = 4.5 \times 15 = 67.5$ N - the $\rho A v^2$ form and the $\dot{m} v$ form agree exactly.

Which flux problem is which

Classification table. Chain lifted: weight + flux; half the flux work heats the pickup. Chain landing: landed weight + flux; reading up to 3x weight. Jet on plate: pure flux $\rho A v^2$; on a moving plate use relative speed. Belt pickup: flux $\dot{m} v$; belt power $\dot{m} v^2$. Rocket: flux $\dot{m} v_e$ = thrust. Rain into cart: retarding flux.

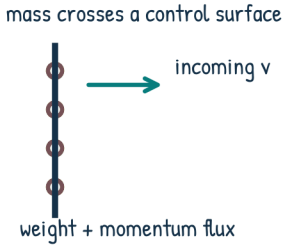
In every case draw the control volume first - what crosses the boundary, at what velocity - then write $\dot{m} \Delta v$.

Which flux problem is which (cont.)

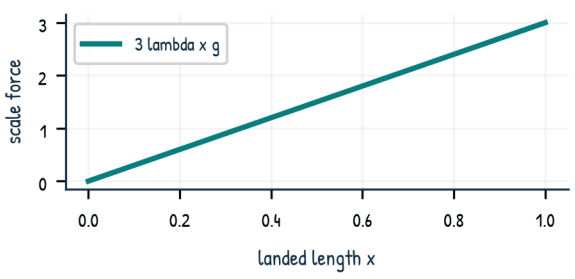
WHY / the control-volume picture is what the fluid-mechanics chapter will formalise as the momentum equation; here it is Newton's second law with a flow diagram.

CHECK / re-derive the universal form: mass dm crosses with velocity u_{in} and leaves/absorbs at u_{out}; force $= \frac{dm}{dt}(u_{out} - u_{in})$ - every entry in the table is this with substituted speeds.

flux: labelled physical setup



flux: limiting curve / diagnostic



REMEMBER / Momentum flux: F = m_dot v = lambda v^2 on top of the carried weight; jets push rho A v^2.

Man on a boat: the COM stays fixed, the parts slide apart

Man on boat and block on wedge; COM-fixed displacements; drills.

Man on a boat: the COM stays fixed, the parts slide apart

System: man (m) + boat (M), initially at rest on frictionless water. No external horizontal force, so the horizontal COM stays FIXED: $mx_m + Mx_b =$ const. If the man walks a distance d RELATIVE to the boat, then $x_m - x_b = d +$ const and the ground displacements satisfy $ms_m + Ms_b = 0$ with $s_m - s_b = d$. Solve:

$s_m = \dfrac{Md}{m+M}$ (man's ground displacement),

$s_b = -\dfrac{md}{m+M}$ (boat slides the other way).

Man on a boat: the COM stays fixed, the parts slide apart (cont.)

Worked (source E8): 60 kg man walks 3 m on a 120 kg boat: $s_m = 120(3)/180 = 2$ m forward; $s_b = -60(3)/180 = -1$ m: the boat moves 1 m opposite. Total check: $60(2) + 120(-1) = 0$ (COM fixed); $s_m + |s_b| = 3 = d$ (the relative walk fully accounted for). Source Q10: 60 kg person JUMPS at 3 m/s relative to the boat:

momentum $0 = 60(3 + v_b) + 120v_b$ (person lab speed $= 3 + v_b$): $180 + 180v_b = 0$, $v_b = -1$ m/s - the boat recoils at 1 m/s.

Man on a boat: the COM stays fixed, the parts slide apart (cont.)

WATCH / Traps: (1) confusing ground displacement with relative displacement - $s_m = Md/(m+M)$ assumes d is measured relative TO THE BOAT; (2) the boat's displacement is opposite; (3) total momentum is zero but the parts' momenta are not; (4) if the water is NOT frictionless, external horizontal friction moves the COM and the fixed-COM shortcut dies.

WHY / the COM line stays vertical-fixed because no horizontal external force acts; every internal shuffle trades equal opposite momenta: $mv_m = -Mv_b$ at every instant; integrate and the displacements follow.

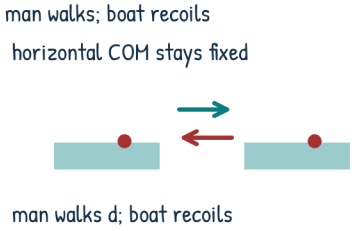
Block on a movable wedge - the same COM law in 2-D

CHECK / source P20: all three statements true: COM fixed; boat opposite; man ground displacement $Md/(m+M)$ with d relative to the boat.

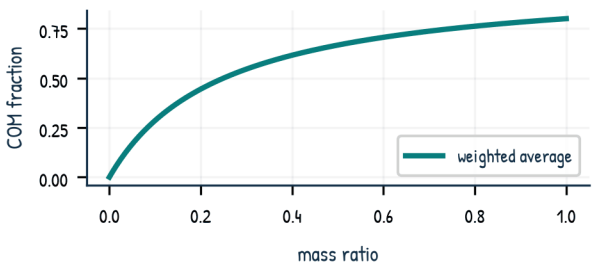
Block (m) slides down a movable wedge (M) on a smooth table: horizontal momentum of block\wedge is conserved (gravity and the table normal are vertical; the block-wedge normal pair is internal). The horizontal COM stays fixed: as the block descends and moves right relative to the wedge, the wedge slides left so that $mx_b + Mx_w =$ const.

Vertical momentum is NOT conserved (the floor's normal force and gravity do not cancel).

boat: labelled physical setup



boat: limiting curve / diagnostic



REMEMBER / Man on boat: horizontal COM fixed; ground displacement = M d/(m + M) opposite to the walk.

Man on a boat: the COM stays fixed, the parts slide apart (cont.)

Man on boat and block on wedge; COM-fixed displacements; drills.

Block on a movable wedge - the same COM law in 2-D (cont.)

Constraint version: if the block slides a relative horizontal distance s_{rel} along the wedge: $s_{block} = \frac{M}{m+M} s_{rel}$, $s_{wedge} = -\frac{m}{m+M} s_{rel}$ - the same formula as the boat. The geometry is identical whenever horizontal external impulse is zero. For a wedge of incline angle α and slide length L along the incline: $s_{rel} = L\cos\alpha$.

Energy: for the wedge problem energy IS conserved (smooth): $mgh = \frac{1}{2}mv_b^2 + \frac{1}{2}Mv_w^2$ with the constraint $v_{bx} = v_{wx} + v_{rel\,x}$ - a two-equation solve. For the boat walk energy is NOT conserved (muscles); use momentum/COM only. The wedge problem is the standard 'both conservation laws, different stages' exam question.

Block on a movable wedge - the same COM law in 2-D (cont.)

CHECK / mass-ratio sanity: $M \gg m$: the wedge barely moves and the block does almost all the sliding; $m \gg M$: the block barely moves and the wedge does almost all the sliding. Both limits pass through the same formula.

(a) 80 kg woman walks 4 m on a 40 kg raft: $s_{raft} = -80(4)/120 = -2.67$ m; $s_{woman} = 1.33$ m.

(b) Two skaters 50 and 70 kg push apart to a relative 3 m/s: $50v_1 = 70v_2$ and $v_1 + v_2 = 3$: $50(3 - v_2) = 70v_2$, $v_2 = 1.25$ m/s, $v_1 = 1.75$ m/s.

(c) Man 60 kg on a 120 kg boat jumps off at 4 m/s relative to the boat:

Internal-motion drills (cont.)

$0 = 60(4 + v_b) + 120v_b$: $v_b = -240/180 = -1.33$ m/s.

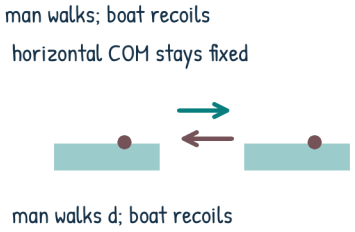
(d) Block 1 kg on a 3 kg wedge slides 0.5 m along the incline (30 deg): relative horizontal $= 0.433$ m; wedge shift $= 1(0.433)/4 = 0.108$ m opposite.

CHECK / drill (c) energy: person KE $= \frac{1}{2}60(2.67^2) = 213$ J; boat KE $= \frac{1}{2}(120)(1.78) = 107$ J; total 320 J = the muscle energy of the jump - momentum zero, energy positive, no contradiction.

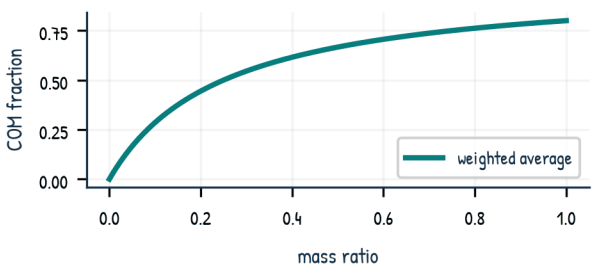
When the COM is NOT fixed: external horizontal forces

If external horizontal forces act (water drag, floor friction, a rope pull), the COM accelerates: $M\mathbf{a}_{cm} = \mathbf{F}_{ext}$. Example: the man-walks-on-boat problem with water drag f opposing the boat: the fixed-COM displacement formula $s_b = -md/(m+M)$ becomes an approximation for short walks;

boat: labelled physical setup



boat: limiting curve / diagnostic



REMEMBER / Man on boat: horizontal COM fixed; ground displacement = M d/(m + M) opposite to the walk.

Man on a boat: the COM stays fixed, the parts slide apart (cont.)

Man on boat and block on wedge; COM-fixed displacements; drills.

When the COM is NOT fixed: external horizontal forces (cont.)

the exact solve is momentum plus the drag impulse. Rule: state the no-external-impulse condition BEFORE using the fixed-COM shortcut.

Drill: woman 80 kg walks 2 m on a 40 kg raft against a constant 12 N drag: during a 1 s walk the COM drifts by $\frac{1}{2}a_{cm}t^2 = \frac{1}{2}(0.1)(1) = 0.05$ m - a small correction, but the exam answer should name it. Drift check: $a_{cm} = 12/120 = 0.1$ m/s^2.

When the COM is NOT fixed: external horizontal forces (cont.)

CHECK / if the drag is proportional to speed, the COM decelerates exponentially (OLY-11 Pattern 1) - the displacement formula then needs the drag integral, not a linear correction.

A ball dropped from h_1 rebounds to h_2 on a fixed floor: impact speed $v = \sqrt{2gh_1}$, rebound speed ev, so $h_2 = e^2 h_1$ and $e = \sqrt{h_2/h_1}$. Source Q5: dropped from 5 m, bounces to 3.2 m: $e = \sqrt{0.64} = 0.8$. Source P10: rebounds to 64%: $e = 0.8$ again.

The height ratio is e^2, never e - the classic trap (drop height 4 m, $e = 0.5$: rebound 1 m, not 2 m).

Bounce heights: e = sqrt(h2/h1) and the geometric ladder (cont.)

Successive bounces (source P25/OL4): $h_n = e^{2n}h_0$ after n bounces. $e = 0.7$ from 10 m: $h_3 = 10(0.7)^6 = 10(0.1176) = 1.176$ m. The heights form a geometric sequence with ratio e^2; the velocities with ratio e; the bounce times with ratio e (since $t = \sqrt{2h/g} \propto e^n$).

Total distance (source P31): $d = h + 2h(e^2 + e^4 + \cdots) = h\frac{1+e^2}{1-e^2}$. For $h = 5$, $e = 0.6$: $d = 5(1.36/0.64) = 10.625$ m. Total time: $t = t_0 + 2t_0(e + e^2 + \cdots) = t_0\frac{1+e}{1-e}$ with $t_0 = \sqrt{2h/g}$: $h = 5$, $g = 10$: $t_0 = 1$ s; $t = 1(1.6/0.4) = 4$ s.

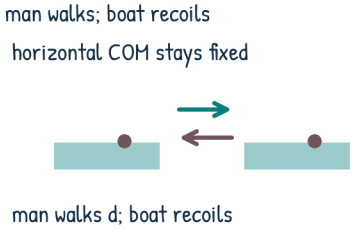
Bounce heights: e = sqrt(h2/h1) and the geometric ladder (cont.)

The two series differ (heights decay by e^2, times by e) - memorise which is which.

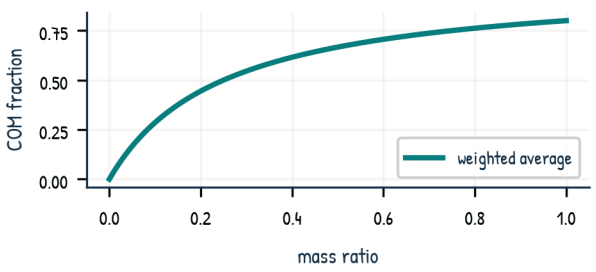
WHY / each bounce is a floor collision with e defined on speeds; $h \propto v^2$ turns the speed ratio e into the height ratio e^2. The geometric series follow because the process is self-similar: every bounce reproduces the previous one scaled.

CHECK / P31(b): how many bounces until $h_n < 0.1$ m from 5 m with $e = 0.6$: $5(0.36)^n < 0.1$, $0.36^n < 0.02$, $n > \ln(0.02)/\ln(0.36) = 3.83$: so after 4 bounces ($h_4 = 5(0.36)^4 = 0.084$ m). The source's printed answer '5' used $|\ln 0.01|$ by mistake; 4 suffices (recorded in SOURCE-AUDIT).

boat: labelled physical setup



boat: limiting curve / diagnostic



REMEMBER / Man on boat: horizontal COM fixed; ground displacement = M d/(m + M) opposite to the walk.

Bounce heights: e = sqrt(h2/h1) and the geometric ladder

e = sqrt(h2/h1); geometric bounce ladder; distance and time series; corrected P31(b); drills.

Series sums and the energy ladder

Energy after n bounces: $E_n/E_0 = h_n/h_0 = e^{2n}$. The energy ladder decays by e^2 per bounce; total energy dissipated to the floor over infinite bounces $= E_0$ (everything eventually). For $e = 0.8$, after 5 bounces: $0.8^{10} = 0.107$ - about 11% of the original energy remains (source OL4).

Distance sum traps: (i) the FIRST drop counts once (h, not $2h$); (ii) each later height counts twice (up and down); (iii) if the ball is thrown DOWN initially, the first impact speed is larger: $v = \sqrt{v_0^2 + 2gh}$ and the first rebound height is $e^2(v_0^2 + 2gh)/2g$. Time sums similar: first drop t_0 once, later flights $2t_0 e^k$.

Series sums and the energy ladder (cont.)

Bounce on a slope: the same e applies to the normal component; the ball hops along the incline with a geometric series in e^2 along the normal - distances along the slope form their own series. Out of scope except olympiad geometry (OLY-04 family).

CHECK / sum check: $d = h\frac{1+e^2}{1-e^2}$ at $e = 0$: $d = h$ (just the drop); at $e \to 1$: $d \to \infty$ (perpetual bouncing). $t = t_0\frac{1+e}{1-e}$ at $e = 0$: $t = t_0$.

Bounce drills

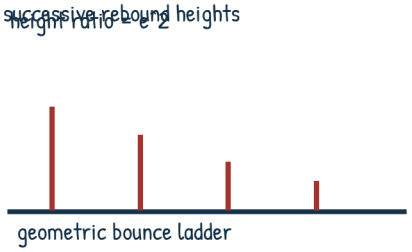
(a) Drop 3.2 m, rebound 1.8 m: $e = \sqrt{0.5625} = 0.75$.
(b) $e = 0.5$ from 8 m: $h_1 = 2$ m, $h_2 = 0.5$ m; $d = 8(1.25/0.75) = 13.33$ m; $t_0 = \sqrt{16/10} = 1.265$; $t = 1.265(1.5/0.5) = 3.8$ s.
(c) $e = 0.9$: after 3 bounces energy $= 0.9^6 = 0.531$ of original - over half remains.

Bounce drills (cont.)

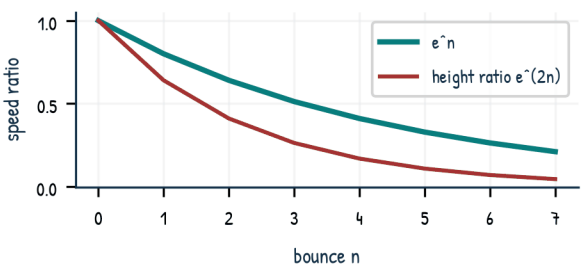
(d) Thrown down at 6 m/s from 5 m: first impact $v = \sqrt{36 + 100} = 11.66$ m/s; with $e = 0.8$ rebound speed 9.33 m/s, height 4.35 m. (e) How many bounces to reach 1% height with $e = 0.7$: $0.49^n = 0.01$: $n = \ln 0.01/\ln 0.49 = 6.46$: 7 bounces.

CHECK / drill (b) distance ladder: $8 + 2(2 + 0.5 + 0.125 + \ldots) = 8 + 2(2.667) = 13.33$ - matches the closed form.

bounces: labelled physical setup



bounces: limiting curve / diagnostic



REMEMBER / Bounce heights fall as e^2n; total distance and time are geometric series in e^2 and e.

Bounce heights: e = sqrt(h2/h1) and the geometric ladder (cont.)

e = sqrt(h2/h1); geometric bounce ladder; distance and time series; corrected P31(b); drills.

Bouncing-ball question types

Type A: find e from heights (one ratio). Type B: find e from a TIME ratio (use $t \propto e^n$). Type C: total distance/time (closed forms). Type D: 'after how many bounces is h below X' (log solve - watch the source's P31 rounding). Type E: energy after n bounces (e^{2n}). Type F: ball on a moving lift/platform (use relative speed in the restitution law - CM-21/OLY-04).

WATCH / Trap: using e instead of e^2 for heights. Trap: including the drop twice in distance sums. Trap: forgetting the rebound time on the way up AND down for intermediate bounces. Trap: applying $e = \sqrt{h_2/h_1}$ when the floor MOVES (then e relates relative speeds, not heights).

Bouncing-ball question types (cont.)

CHECK / self-consistency: from h_1, h_2 alone compute both e and the ratio $h_3 = e^2 h_2$; if a problem gives three heights, they must satisfy $h_2^2 = h_1 h_3$ (geometric middle) - e.g. Q23's 10, 6.4, 4.1: $6.4^2 = 40.96$ vs $10 \times 4.1 = 41$ (rounding: $e = 0.8$ exactly gives 4.096).

A wall moves at speed v_w; a ball with velocity v_{before} (signed, same axis) bounces with restitution e measured in RELATIVE speeds: $v_{after} - v_w = -e(v_{before} - v_w)$. Solve: $v_{after} = v_w - e(v_{before} - v_w)$. In the wall's frame it is an ordinary bounce (relative velocity reverses, scaled by e); transform back by adding v_w.

Bouncing off a MOVING wall: the wall's speed enters the restitution.

Worked (source E9, with the source's own check-block error corrected): ball at 10 m/s toward a wall coming toward it at 5 m/s, $e = 0.8$. Take the ball's direction positive: $v_{before} = +10$, $v_w = -5$. Relative approach: $10 - (-5) = 15$ m/s. $v_{after} = -5 - 0.8(15) = -17$ m/s:

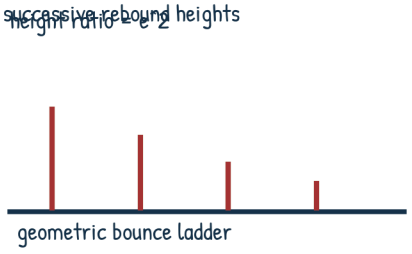
the ball rebounds at 17 m/s - FASTER than it came in, because the wall's KE paid the difference. The source's 'Check' block says 7 m/s (a sign slip: it used $v_{after} - v_w = +12$ instead of -12); the source's own solution block reaches the correct 17 m/s. Recorded in SOURCE-AUDIT.

Bouncing off a MOVING wall: the wall's speed enters the restitution.

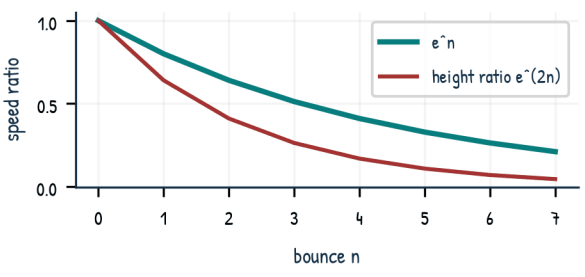
Source Q14: 1 kg ball at 3 m/s toward a wall moving toward it at 2 m/s, $e = 0.5$: $v_{after} = -2 - 0.5(5) = -4.5$ m/s: rebounds at 4.5 m/s. General lesson: a wall approaching at u adds roughly $(1 + e)u$ to the rebound speed; a receding wall subtracts it. Superball limit ($e = 1$, wall speed u toward a ball at v): rebound $= v + 2u$ - the famous 'bounce off a moving train' boost.

WHY / restitution measures the relative speed of separation over approach AT THE CONTACT. The wall's motion changes both. Applying e to lab speeds is the classic error - the formula $v_{after} - v_w = -e(v_{before} - v_w)$ encodes the frame change in one line.

bounces: labelled physical setup



bounces: limiting curve / diagnostic



REMEMBER / Bounce heights fall as e^2n; total distance and time are geometric series in e^2 and e.

Bouncing off a MOVING wall: the wall's speed enters the restitution law

Moving-wall formula with the source's E9 check-block corrected; platform energy pump; drills.

Bouncing off a MOVING wall: the wall's speed enters the restitution.

CHECK / E9 energy sanity (1 kg ball): KE grew from $\frac{1}{2}(100) = 50$ J to $\frac{1}{2}(289) = 144.5$ J - no violation; the wall's motor paid. Q14: $\frac{1}{2}(9) = 4.5$ J to $\frac{1}{2}(20.25) = 10.125$ J - the approaching wall did work.

A ball bounces on a vertically moving platform (full resonance story in OLY-04). One bounce, up-positive: ball at v_{before} (negative if falling), platform at v_w (positive if rising):
$v_{after} = v_w - e(v_{before} - v_w) = (1+e)v_w - ev_{before}$
. Platform rising at u meets ball falling at speed v :

Moving platform (the energy-pump machine) (cont.)

$v_{after} = (1+e)u + ev$ upward - larger than the fixed-floor rebound ev : the platform pumped $(1+e)u$ into the ball in one bounce.

Receding platform (falling away at u): the rebound weakens the same way. The invariant recipe: relative speeds in the e law, then transform back; never apply e to lab speeds when the surface moves. Elastic platform limit ($e = 1$, rising at u): $v_{after} = 2u + v$ - the ball leaves with its own speed plus twice the platform's.

CHECK / limit $v_w = 0$: reduces to the fixed-floor law $v_{after} = -ev_{before}$. Elastic head-on ball vs rising platform: relative approach $v + u$; relative separation $v + u$; lab $= (v+u) + u = v + 2u$ - superelastic; platform energy transferred.

Wall-collision drills

(a) Ball at 12 m/s hits a wall receding at 3 m/s (same direction as the ball), $e = 0.9$:
$v_{after} = 3 - 0.9(12 - 3) = 3 - 8.1 = -5.1$ m/s: rebounds at 5.1 m/s - the receding wall 'stole' speed. (b) Ball at 8 m/s meets a wall coming at 6 m/s, $e = 1$:
$v_{after} = -6 - 1(8 + 6) = -20$ m/s: rebounds at 20 m/s.

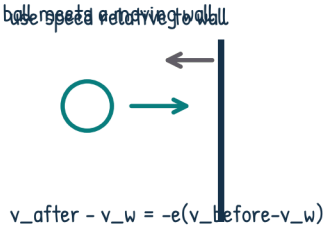
(c) Ball dropped 1.25 m ($v_{before} = -5$ m/s) onto a platform rising at +2 m/s, $e = 0.5$:
$v_{after} = 2 - 0.5(-5 - 2) = 2 + 3.5 = 5.5$ m/s upward - use the formula with signed inputs, not ad-hoc speed arithmetic.

Wall-collision drills (cont.)

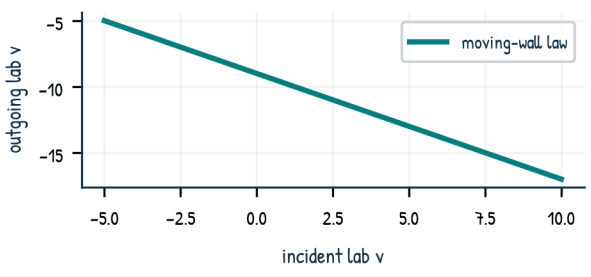
CHECK / drill (c) energy audit: ball KE in $= \frac{1}{2}m(25)$; out $= \frac{1}{2}m(30.25)$ - the ball gained, the rising platform did work. Drill (b): 32 J to 200 J for 1 kg - the moving wall paid.

A ball bouncing between two walls (one or both moving): apply $v' = v_w - e(v - v_w)$ per contact, alternating walls. Between two walls approaching at u each (ball speed v in lab): each bounce adds roughly $(1+e)u$ - the ball heats up like a gas piston; between two receding walls it cools. The wall-frame composition is the olympiad 'bouncing ball in a closing gap' family (OLY-04's fixed-platform cousin).

wall: labelled physical setup



wall: limiting curve / diagnostic



REMEMBER / Moving wall: v_after - v_w = -e (v_before - v_w); a wall coming at you can double the rebound speed.

Bouncing off a MOVING wall: the wall's speed enters the restitution law (cont.)

Moving-wall formula with the source's E9 check-block corrected; platform energy pump; drills.

Bouncing between two walls: compose wall frames (cont.)

Drill: ball at 8 m/s between a fixed wall and a wall approaching at 3 m/s, $e = 1$: first bounce off the moving wall: $v' = -3 - 1(8 + 3) = -14$ m/s toward the fixed wall; off the fixed wall: $+14$; back at the moving wall at 14: $v'' = -3 - 1(14 + 3) = -20$ m/s - grows by 6 m/s per round trip.

CHECK / energy audit per round trip: the approaching wall does work on the ball each bounce; the ball's KE grows - no conservation law is broken (the wall is external and driven).

The ten examiner traps, sharpened

(1) Energy conservation across an INELASTIC collision - never; use momentum (and e if given). (2) Applying e to the full velocity - e lives only along the line of impact; tangential components are untouched for smooth bodies. (3) Forgetting the exhaust momentum in rocket problems - the rocket equation IS the exhaust ledger. (4) Using v_{cm} as the velocity of every part after impact - only stuck parts share v_{cm}; separated parts do not.

(5) Mixing system and particle laws - $Ma_{cm} = F_{ext}$ for the system, $F = ma$ for each piece.

The ten examiner traps, sharpened (cont.)

(6) Assuming the COM stays fixed when external forces act - fixed only when $\int F_{ext}\,dt = 0$ in that direction. (7) Confusing impulse (N s) with force (N) - $J = \bar{F}\Delta t$. (8) Forgetting momentum is a vector - components conserve independently. (9) 'Heavy ball bounces back' - no: the heavy plows on, the light flies at nearly $2u_1 - u_2$.

(10) Using the ballistic-pendulum formula when the bullet EXITS - that formula assumes embedding ($e = 0$).

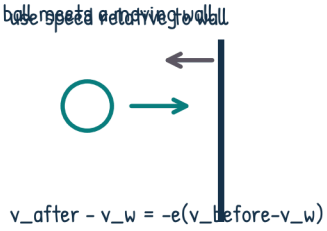
WHY / every trap is a broken conservation statement or a broken scope condition. The pre-submission audit (CM-23) checks exactly these.

Trap 11 (source C9, corrected): equal bounce angles are NOT free

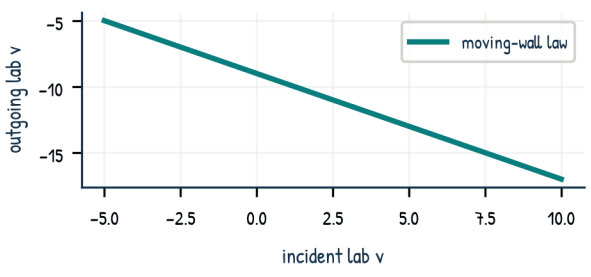
Source C9 asks: a ball hits a wall at angle θ and bounces at angle θ - is the collision elastic? The source answers 'not necessarily: the angle of reflection equals the angle of incidence for any smooth-sphere collision regardless of e'. That is WRONG for $e < 1$. Correct physics: for a smooth wall the tangential component v_t is unchanged and the normal component becomes ev_n. The rebound angle from the normal:

$\tan\theta' = v_t/(ev_n) = (\tan\theta)/e > \tan\theta$ for $e < 1$ - the ball leaves MORE GRAZING, not at the mirror angle. Equal angles then imply $e = 1$ exactly (smooth wall, no spin). With friction (rough wall or spinning ball) the tangential impulse also changes v_t and equal angles no longer pin down e - only then is 'not necessarily' fair.

wall: labelled physical setup



wall: limiting curve / diagnostic



REMEMBER / Moving wall: v_after - v_w = -e (v_before - v_w); a wall coming at you can double the rebound speed.

The ten examiner traps, sharpened

Ten standard traps; the wall-angle correction (C9); the 1-D exchange scope (P5); sign and arithmetic discipline.

Trap 11 (source C9, corrected): equal bounce angles are NOT free (.

Numbers: impact at 45 degrees from the normal at 10 m/s, $e = 0.5$: $v_t = v_n = 7.07$; the normal part becomes 3.54: rebound angle $\arctan(7.07/3.54) = 63.4$ degrees from the normal - far from 45. Energy: $K_f/K_i = \sin^2 45 + e^2\cos^2 45 = 0.625$ - visibly not elastic.

CHECK / limit $e = 1$: $\tan\theta' = \tan\theta$ - specular reflection. Limit $e \to 0$: $\theta' \to 90$ degrees - the ball skims the wall (normal motion killed, tangential survives). The formula interpolates physically.

Trap 12 (source P5 ambiguity): velocity exchange is 1-D only

Source P5: 'two equal-mass balls collide elastically, one at rest; after: (a) the incoming ball stops (d) they move at 90 degrees' with answer (a). Correct only in 1-D. In 2-D the general outcome is the 90-degree separation with the shooter deflected at speed $u\cos\theta_1$ - it stops only in the head-on limit. Safe exam sentence: 'in 1-D the velocities exchange (shooter stops);
in 2-D they separate at 90 degrees and the shooter generally keeps speed $u|\cos\theta_1|$'.

Trap 12 (source P5 ambiguity): velocity exchange is 1-D only (cont.

WATCH / Trap 13 (source Q13 under-specification): 'a chain of mass 3 kg is lifted at 2 m/s, find the force' has no unique number without a length scale - the force grows with the hanging length: $F = \lambda x g + \lambda v^2$. Answer with the formula (as source P34c does) and define x and $\lambda = M/L$.

WHY / ambiguity traps are scope traps: the statement leaves a free parameter (dimension, length, frame). Name the parameter and answer parametrically - examiners award the method.

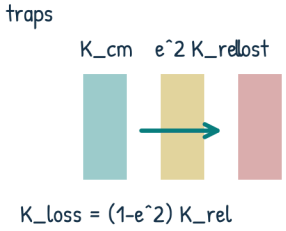
CHECK / trap 12 numbers: equal masses, shooter at u, scatter angle 30: shooter speed $0.866u$ (NOT zero) - one number kills the 'always stops' reading.

Sign traps and arithmetic traps

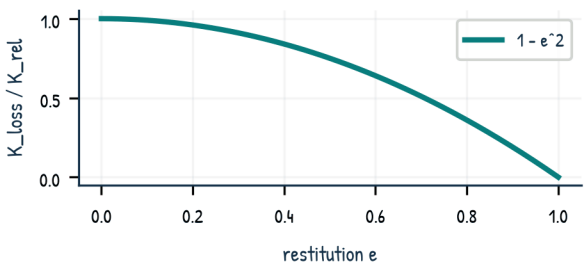
Sign trap: $e = (v_2 - v_1)/(u_1 - u_2)$ needs consistent directions; in 2-D project first. Sign trap: the wall formula needs signed velocities; write the axis before numbers. Sign trap: $K_{loss} = K_i - K_f$ comes out negative for superelastic - not an error, a released spring.

Arithmetic trap (documented in SOURCE-AUDIT): in $v_1 = \frac{(m_1 - em_2)u_1 + (1+e)m_2u_2}{m_1 + m_2}$ the factor $(1+e)$ multiplies the WHOLE m_2u_2 term. One dropped $\frac{1}{2}$ in $\frac{1}{2}(3)v_2^2$ produced the source's Q18 error ($6/7$ instead of $6/5$ for v_1^2). Three-invariant audits (momentum, e, energy) catch every such slip.

traps: labelled physical setup



traps: limiting curve / diagnostic



REMEMBER / Traps: e only along the line of impact; equal bounce angles imply e = 1 on a smooth wall; exchange is 1-D only.

The ten examiner traps, sharpened (cont.)

Ten standard traps; the wall-angle correction (C9); the 1-D exchange scope (P5); sign and arithmetic discipline.

Sign traps and arithmetic traps (cont.)

CHECK / re-run one old trap against the ledger: 'all KE is lost when things stick'? - NO: $K_{cm} = \frac{1}{2}Mv_{cm}^2$ survives; equal masses lose exactly half.

'Find the COM' -> definition or negative-mass trick (CM-01/02). 'Explosion / recoil / fragments' -> COM continues, momentum for the pieces (CM-03/05). '1-D collision' -> momentum + e, or the master formulas (CM-07/08). '2-D collision / oblique' -> resolve along the line of impact (CM-11/12). 'Ballistic pendulum / bullet-block' -> momentum first, energy second (CM-13).

Triage: route by keyword before touching a number (cont.)

'Bullet passes through' -> momentum only for the contact (CM-13). 'Spring compression' -> stage bookkeeping then $\frac{1}{2}kx^2$ (CM-14).

'Rocket / variable mass / jet / belt / chain' -> flux law $\dot{m}\Delta v$ or the rocket equation (CM-16/17/18). 'Man on boat / block on wedge' -> COM fixed in the free direction (CM-19). 'Bounce heights' -> $e = \sqrt{h_2/h_1}$ ladder (CM-20). 'Moving wall / moving platform' -> relative-speed restitution (CM-21). 'Bouncing between walls' -> wall frames composed (CM-21).

Triage: route by keyword before touching a number (cont.)

'Maximum / minimum speed after' -> CM-frame circle locus (CM-15/OLY-02).

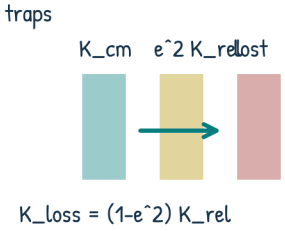
WHY / triage prevents the master sin: applying the wrong conservation law. The keyword names the stage structure; the stage structure names the laws. Every problem in the chapter is one to three stages of (momentum, energy, restitution, flux).

Validity map: when each formula breaks

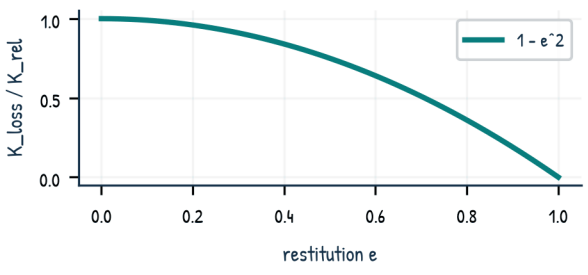
$\mathbf{R} = \sum m_i \mathbf{r}_i / M$: always (definition); continuous bodies need integration. $M\mathbf{a}_{cm} = \mathbf{F}_{ext}$: always in Newtonian mechanics (needs third-law pairs). $\sum \mathbf{p} =$ const: when net external impulse vanishes in that direction - brief collisions OK by approximation; long pushes not. $e = v_{sep}/v_{app}$:

along the line of impact, passive contacts; superelastic allowed with $e > 1$. Master formulas (3.7): 1-D or normal-direction only; fixed masses. $K_{loss} = \frac{1}{2}\mu u_{rel}^2(1 - e^2)$: same scope; negative means energy gained.

traps: labelled physical setup



traps: limiting curve / diagnostic



REMEMBER / Traps: e only along the line of impact; equal bounce angles imply e = 1 on a smooth wall; exchange is 1-D only.

Triage: route by keyword before touching a number

Triage tree; validity map; pre-submission audit and timing.

Validity map: when each formula breaks (cont.)

$\Delta v = v_e \ln(m_0/m_f)$: constant v_e, no external forces (add $-gt_b$ with gravity); collinear exhaust. $v_{after} - v_w = -e(v_{before} - v_w)$: one-dimensional wall contact; smooth. Boat/wedge COM laws: zero external impulse in the tracked direction; frictionless supports. 90-degree rule: equal masses, elastic, smooth, target at rest. Bounce ladder: fixed floor, constant e.

Pre-submission audit (ten points)

CHECK / before using any formula, say its validity sentence out loud. If a condition fails, fall back to first principles: momentum ledger + energy ledger + geometry.

(1) COM: correct reference point and signed masses. (2) Momentum: asserted why net external impulse is zero (in the relevant direction). (3) Collision classified: elastic / inelastic / sticking, correct e. (4) e applied along the line of impact only. (5) Two-stage problems: momentum first, energy second (order locked). (6) Rocket: $\ln(m_0/m_f)$, not $m_0 - m_f$; v_e relative. (7) 2-D:

Pre-submission audit (ten points) (cont.)

decomposed along and across the line of impact. (8) Energy loss: μ and u_{rel}, not raw masses. (9) Units consistent; $g = 9.8$ or 10 as the problem states. (10) Every sub-part answered; signs of final velocities stated as directions.

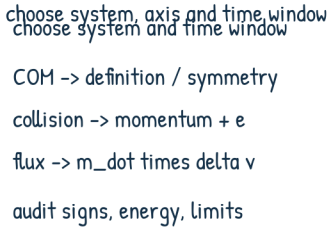
Time plan for a 3-question collision section: 2 min triage, 3 min per 1-D problem, 5 min for 2-D or two-stage, 2 min audit. COM problems die by definition; collision problems die by momentum + e; flux problems die by $\dot{m}\Delta v$.

Pre-submission audit (ten points) (cont.)

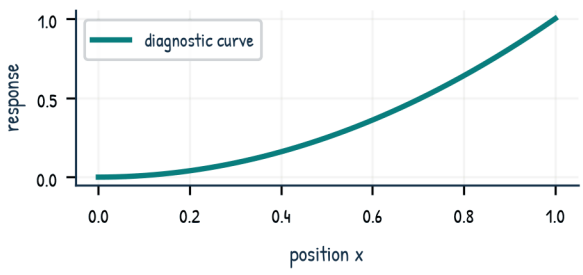
CHECK / the audit's favourite catch: a final velocity reported as a speed when the problem asked for velocity - state direction words ('rebounds at 17 m/s opposite the incoming direction').

When a collision involves a constraint (ball on a string, rod, hinge), the constraint delivers an IMPULSE too: split the impulse into the contact part and the constraint part, or choose coordinates where the constraint impulse does no work (along the string). Example: ball on a string strikes a block: during the impact the string may go slack or jerk - the safe exam route:

route: labelled physical setup



route: limiting curve / diagnostic



REMEMBER / Triage first: COM definition, explosion rides the COM, collisions are momentum + e, stages never mix their laws.

Triage: route by keyword before touching a number (cont.)

Triage tree; validity map; pre-submission audit and timing.

Mixed-constraint problems: collision plus string, rod or hinge (con.

momentum of the SYSTEM (ball\block) along the free direction, plus e along the line of impact, plus the string condition (tension ≥ 0).

Hinge problems: angular momentum about the hinge is conserved during the collision (the hinge impulse has zero moment arm); linear momentum is not. Ball hitting a hanging rod: $L = I\omega$ about the pivot, then energy swings the rod - the same two-stage protocol with I in place of m .

Mixed-constraint problems: collision plus string, rod or hinge (con.

CHECK / the audit: count unknowns (post velocities, constraint impulses) and equations (momentum, e , constraint kinematics); if the count mismatches, a physical condition (slack string, hinge direction) is missing.

$\mathbf{R} = \frac{1}{M}\sum m_i \mathbf{r}_i$ and $M\mathbf{a}_{cm} = \mathbf{F}_{ext}$ (4.1) - the definition and the only equation of motion for a system.

$\mathbf{J} = \Delta\mathbf{p} = \int \mathbf{F}\,dt$ (4.2) - impulse equals momentum change. $\sum \mathbf{p}_{before} = \sum \mathbf{p}_{after}$ when $\sum \mathbf{F}_{ext} = 0$ (4.3) - conservation, component-wise.

The boxed ledger: six formulas that run the chapter (cont.)

$e = \dfrac{v_{sep}}{v_{app}} = \dfrac{v_2 - v_1}{u_1 - u_2}$ (4.4) - restitution along the line of impact. $K_{loss} = \frac{1}{2}\mu(u_1 - u_2)^2(1 - e^2)$ (4.5) - the energy price of any 1-D collision. $\Delta v = v_e \ln \frac{m_0}{m_f}$ (4.6) - the rocket. Plus the two workhorses: $v_{after} - v_w = -e(v_{before} - v_w)$ (moving wall) and $K = \frac{1}{2}Mv_{cm}^2 + \frac{1}{2}\mu v_{rel}^2$ (CM decomposition).

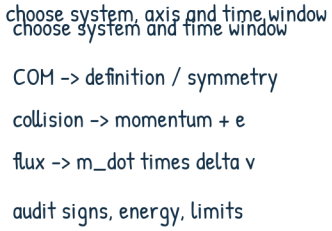
WHY / the ledger is deliberately short: everything else in the chapter is a derived special case or a bookkeeping protocol. If a problem does not reduce to one of these plus geometry, the triage was wrong.

Limits checklist (each formula, one limit)

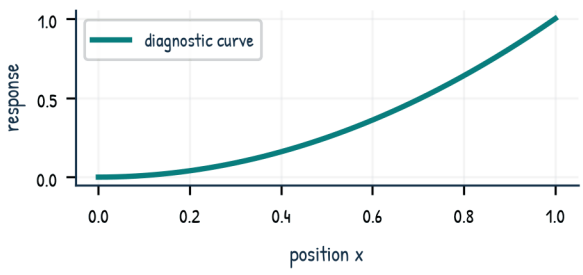
(4.1): $n = 1$ gives $\mathbf{R} = \mathbf{r}_1$. (4.2): constant force gives $J = F\Delta t$. (4.3): switch off external force and v_{cm} freezes. (4.4): $e = 0$ sticks, $e = 1$ elastic. (4.5): $e = 1$ gives zero loss; $e = 0$ gives $\frac{1}{2}\mu u_{rel}^2$. (4.6): $m_f = m_0$ gives $\Delta v = 0$; $m_0/m_f = e$ gives $\Delta v = v_e$ (63% of the mass is propellant).

Which formula when: mass distribution -> (4.1) or negative mass. Force and time -> (4.2). Collision without external force -> (4.3) + (4.4). Given e and velocities -> energy lost from (4.5). Rocket -> (4.6). Pendulum -> (4.3) then energy. Chain/jet/belt -> $\dot{m}\Delta v$ flux. Boat/wedge -> fixed-COM displacements. Bounce heights -> $e = \sqrt{h_2/h_1}$.

route: labelled physical setup



route: limiting curve / diagnostic



REMEMBER / Triage first: COM definition, explosion rides the COM, collisions are momentum + e, stages never mix their laws.

The boxed ledger: six formulas that run the chapter

Boxed formula ledger; limits; which-formula-when; derivation habit.

Limits checklist (each formula, one limit) (cont.)

CHECK / final consistency: any collision answer must satisfy three audits - signed momentum, e -check, and the energy ledger (surviving $K_{cm} + e^2 K_{rel}$). Any COM answer must shift correctly under a change of origin.

(a) 2 kg at 5 into 2 kg at rest elastic: $v_1 = 0$, $v_2 = 5$ - ledger (4.4) with $m_1 = m_2$. (b) $e = 0.7$ bounce: height ratio 0.49. (c) μ for 3 kg and 6 kg: 2 kg. (d) Rocket 3 km/s exhaust, mass ratio e: $\Delta v = 3$ km/s. (e) Impulse to stop 0.15 kg at 25 m/s: 3.75 N s. (f) Explosion of 8 kg at rest into 5 and 3: 5 kg piece at 3 m/s east -> 3 kg piece at 5 m/s west.

One-minute drills on the ledger (cont.)

Two-minute version: derive (4.5) from the decomposition; derive (4.6) from momentum in dt; prove the 90-degree theorem with a dot product. These three derivations are the olympiad floor and the JEE ceiling - if you can rebuild the ledger, you do not need to memorise it.

CHECK / the derivation habit: every boxed formula above appears in the sheets as a derivation with its assumptions. The ledger is the index; the cards are the proof.

Rebuild-the-formula drills (60 seconds each)

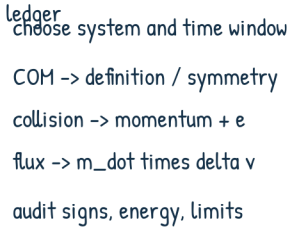
(a) Derive $K_{loss} = \frac{1}{2}\mu u_{rel}^2(1 - e^2)$ from $K = \frac{1}{2}Mv_{cm}^2 + \frac{1}{2}\mu v_{rel}^2$ (CM-09). (b) Derive $\Delta v = v_e \ln(m_0/m_f)$ from $m\,dv = -v_e dm$ (CM-17). (c) Prove the 90-degree rule with $\mathbf{u} = \mathbf{v}_1 + \mathbf{v}_2$ and $u^2 = v_1^2 + v_2^2$ (CM-12). (d) Derive the hole trick $X = \frac{M_1 r_1 - M_2 r_2}{M_1 - M_2}$ from negative mass (CM-02).

Rebuild-the-formula drills (60 seconds each) (cont.)

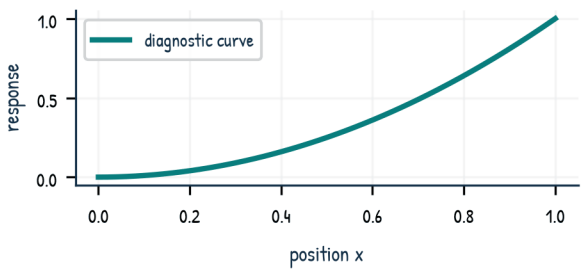
Each derivation is one algebra chain plus one stated assumption: (a) momentum + e; (b) momentum in dt, constant v_e; (c) momentum + energy + equal masses; (d) superposition of a full body and a negative-mass plug. If you cannot rebuild one in a minute, re-read that sheet's first card.

CHECK / the four derivations span the whole ledger: definition (4.1) used in (d), impulse (4.2) in (b), conservation (4.3) in (a) and (c), restitution (4.4) in (a) and (c). Nothing in the chapter outruns them.

ledger: labelled physical setup



ledger: limiting curve / diagnostic



REMEMBER / The ledger: six boxed formulas, their limits, and the one table that says which formula when.